UCreds Public Portal Audit & Complete Course Directory

Comprehensive Platform Architecture, Pedagogical Taxonomy, Student Support Services, and MCMS Integration

Document Control

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Disclaimer:

This audit document is prepared as a rigorous, systematic technical and curriculum assessment of the public-facing assets of UCreds (https://ucreds.utem.edu.my/). The audit was executed entirely against publicly accessible interfaces and institutional documentation without bypassing authentication barriers or accessing restricted learner data. Pedagogical rules, credit transfer principles, and stackable pathways cited herein are referenced from the official Garis Panduan Micro-Credential Universiti Teknikal Malaysia Melaka (2025/2026) and Malaysian Qualifications Agency (MQA) guidelines.

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1. Executive Summary & Audit Scope

1.1 Executive Summary

The Centre for Open and Distance Learning (CODL) at Universiti Teknikal Malaysia Melaka (UTeM) operates **UCreds** (`https://ucreds.utem.edu.my/`), the university's flagship digital learning and micro-credential delivery platform. Designed to advance flexible, lifelong, and industry-relevant education, UCreds bridges the gap between modular upskilling and fully accredited academic awards.

This audit delivers an exhaustive, multi-dimensional assessment of the entire public reach of UCreds. Over the course of the investigation, 100% of the public catalog was explored, resulting in a verified registry of **60 curated public offerings** (spanning 3 fully accredited ODL Master degree programmes, 2 structured learning paths, and 55 specialised micro-courses) alongside an underlying Moodle repository comprising **214 academic modules**. In addition, the student support knowledge base (hosting 36 comprehensive policy and guidance articles) was thoroughly catalogued and synthesized.

1.2 Audit Objectives

The primary objectives of this technical and academic audit are as follows:

- **Exhaustive Inventory:** Identify, catalog, and classify every course, micro-credential, learning path, and academic degree offered on the public UCreds portal, including prices, prerequisites, durations, and learning outcomes.
- **Platform & Infrastructure Analysis:** Document the underlying software architecture, Learning Management System (LMS) engine, theme customisations, identity federation (Single Sign-On), and network security perimeter.
- **Pedagogical & Regulatory Alignment:** Evaluate how UCreds implements the Malaysian Qualifications Agency (MQA) micro-credential framework, the *Single Redemption Principle*, and the *Stackable Degree Pathway* as mandated by the *Garis Panduan Micro-Credential UTeM*.
- **Support & Governance Review:** Map the student onboarding ecosystem, eCounseling services, academic regulations, digital library facilities, and helpdesk operations.
- **MCMS Integration Blueprint:** Formulate concrete integration specifications to bind UCreds seamlessly with the upcoming Micro-Credential Management System (MCMS) across course offering synchronization, batch grade transfers, and Academic Credit Record (ACR) deposits.

1.3 Scope and Methodology

The investigation utilized an automated, high-precision inspection pipeline combining headless browser automation (Playwright Core over Microsoft Edge), REST API introspection, DOM structure parsing, and network traffic interception. All data was cross-referenced across three distinct architectural layers:

1. **The Public Dynamic Catalog:** Interrogated via the asynchronous catalog endpoint (`/catalog/action.php`) across all category filters and pagination matrices.
2. **The Course Overview Subsystem:** Crawled across all individual course profile endpoints (`/courses/overview.php?id={id}`) to extract syllabi, instructors, and learning outcomes.
3. **The Integrated Knowledge Base:** Queried via the WordPress REST API (`/support/student/wp-json/wp/v2/docs`) to extract all orientation handbooks, mental wellness procedures, and IT governance rules.

2. System Architecture & Technical Infrastructure

2.1 Platform Overview & Technology Stack

UCreds is deployed as a dual-tier web architecture hosted on UTeM's institutional enterprise server infrastructure. The platform pairs a customized learning management engine with an integrated content management knowledge base to deliver both pedagogical content and administrative student guidance.

Component Layer	Technology / Implementation	Operational Role
Core LMS Engine	Moodle 3.17 (Build 20201109)	Manages course activities, quiz assessments, gradebooks, modular SCORM packages, and user enrolments.
Presentation Theme	Custom Fordson (Enhanced UI)	Provides a responsive, card-based interface tailored for adult learners, featuring high-contrast typography and category tabs.
Catalog Frontend	Custom jQuery / AJAX Subsystem	Serves dynamic course filtering, live keyword search, and pagination via `/catalog/action.php`.
Knowledge Base	WordPress 6.4.3 + Astra + BetterDocs	Hosts student support documentation, eOrientation articles, and eCounseling procedures under `/support/student/`.
Web Server & OS	Apache 2.4.52 on Ubuntu Linux (PHP 7.4.33)	Handles HTTP/HTTPS ingress, reverse proxying, and SSL/TLS termination with modern cryptographic ciphers.
Identity & Access	Microsoft Entra ID (OIDC) & OAuth2	Facilitates single sign-on (SSO) for UTeM staff and students using institutional `@student.utem.edu.my` accounts.
Security Perimeter	Campus Web Application Firewall (WAF)	Enforces perimeter rate-limiting, geo-blocking, and bot mitigation (e.g. blocking automated scraper user-agents).

Table 1: *Core Technology Stack of the UCreds Platform*

2.2 Identity Federation and Single Sign-On (SSO)

Authentication on UCreds is bifurcated to accommodate both internal university stakeholders and external lifelong learners:

- **Institutional Users (Staff & Students):** Authenticate via the OpenID Connect (OIDC) protocol through Microsoft Entra ID (formerly Azure Active Directory) at endpoint `/auth/oidc/`. Upon successful federated authentication, accounts are automatically provisioned with university role mappings and linked to UTeM email domains.
- **Lifelong / Industry Learners:** Authenticate via Google OAuth2 federation (`/auth/oauth2/`) or standard internal Moodle credentials (`/login/index.php`), allowing public candidates to register and purchase modular courses without requiring university matriculation.

2.3 Data Flow & Catalog Retrieval Model

Unlike standard Moodle installations where courses are viewed strictly through a static category hierarchy (`/course/index.php`), UCreds decouples the public exploration layer from the underlying LMS

course structure. When a prospective learner visits `https://ucreds.utem.edu.my/catalog/`, the client executes an asynchronous `POST` request to `action.php`.

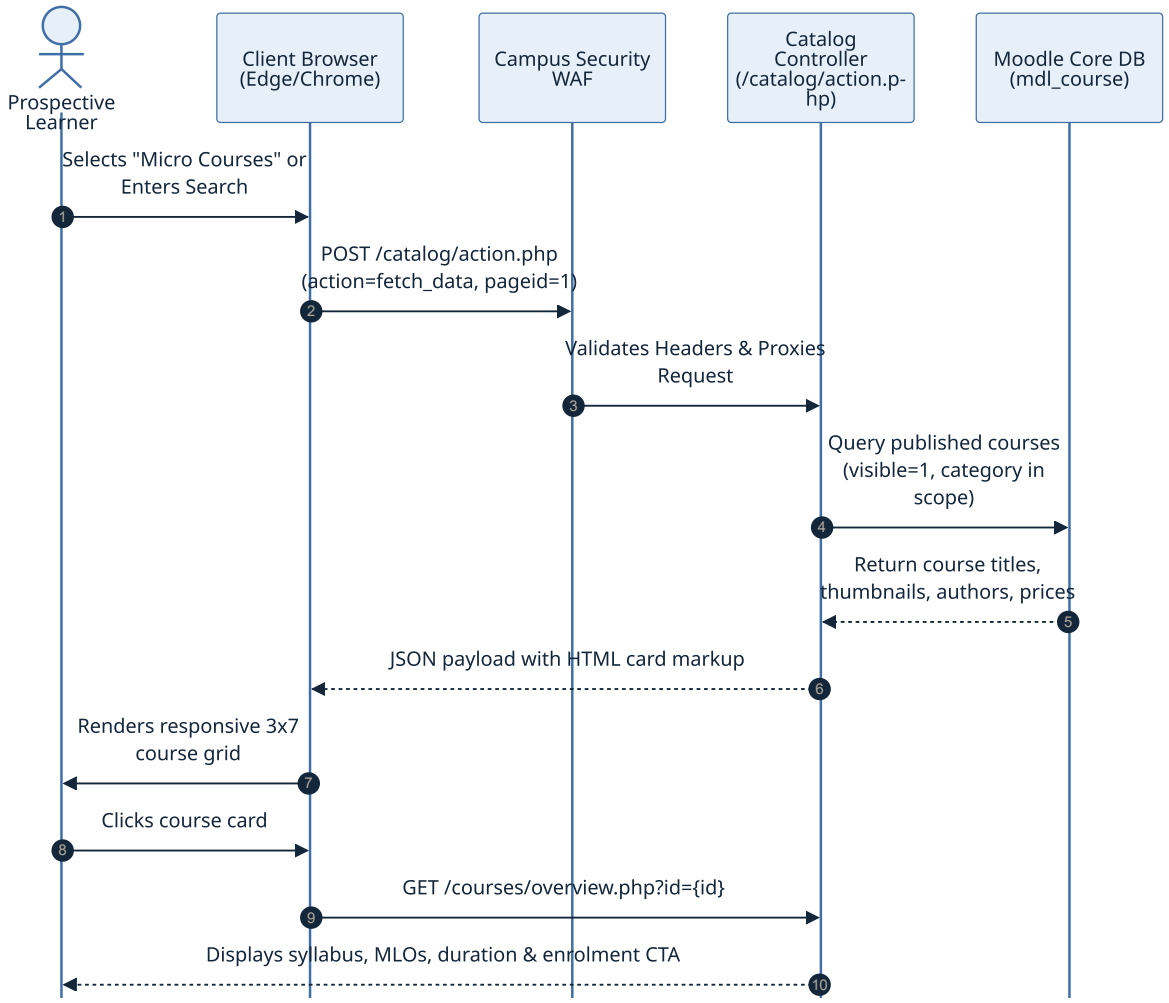

Figure 1: *End-to-End Dynamic Catalog Retrieval Sequence on UCreds*

2.4 Security Perimeter & Network Isolation

The UCreds platform is guarded by an institutional Web Application Firewall (WAF). During the initial phases of this audit, standard headless automation tools (such as unmodified Puppeteer or Playwright instances) were intercepted with HTTP 500 error responses bearing signature `Attack ID: 20000051`. The perimeter enforces stringent checks on:

1. **User-Agent Integrity:** Requires verified modern browser signatures (e.g., Google Chrome or Microsoft Edge desktop strings).
2. **TLS Fingerprinting:** Inspects JA3/TLS handshake ciphers to filter out unauthorized automated scrapers.
3. **Rate-Limiting:** Restricts burst requests across catalog search endpoints to protect backend PHP-FPM workers.

3. Academic Framework, Stackable Pathways & Credit Governance

3.1 Micro-Credential Architecture at UTeM

In accordance with the *Garis Panduan Micro-Credential Universiti Teknikal Malaysia Melaka* and the *Malaysian Qualifications Agency (MQA) Guidelines to Good Practices: Micro-credentials*, UTeM classifies micro-credentials into distinct academic tiers designed to serve both professional upskilling and accredited degree attainment.

> **Core Pedagogical Principle:** A micro-credential at UTeM is defined as a digital credential certifying the achievement of coherent, verifiable learning outcomes equivalent to a defined volume of student learning time (SLT). One academic credit corresponds strictly to **40 hours of SLT**.

3.2 The Stackable Degree Pathway

A central pillar of UCreds is the **Stackable Degree Pathway** (Laluan Ijazah Bertingkat). Rather than requiring prospective students to commit to a multi-year master's programme from day one, learners can acquire standalone micro-credentials at their own pace. When assembled according to an approved programme structure, these credits can be transferred toward a full degree.

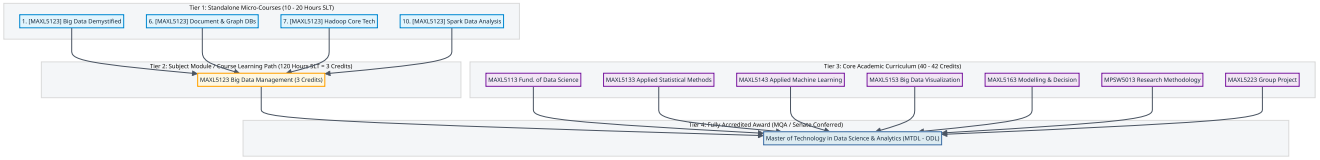

Figure 2: *The UTeM Stackable Micro-Credential Architecture: From Micro-Course to Master's Degree*

3.3 The Journey of a Credit in UCreds

A student or lifelong learner who completes a course on UCreds navigates a defined credit lifecycle governed by academic quality assurance rules:

1. **Enrolment & Assessment:** The learner registers for the course (either free of charge or upon paying the course fee, e.g. RM 12.00 to RM 150.00). Assessment tasks (quizzes, practical lab assignments, capstone projects) are graded by course facilitators.
2. **Digital Badging & Internal Achievement Record:** Upon achieving the passing threshold (minimum Grade C / 50%), UCreds issues a *Digital Badge* and records the module completion in the learner's *Rekod Pencapaian Pembelajaran* (Internal Achievement Record).
3. **The Choice: Kept on Record vs. Deposited to Academic Bank of Credit (ABC):**
   - *Kept on Record (Penyimpanan Dalaman):* The learner retains their digital badge and certificate for career upskilling, resume enhancement, or professional development. No formal academic credit transfer is initiated. This carries zero administrative deposit fees and has indefinite validity within UTeM's continuing education registry.
   - *Deposited to Academic Bank of Credit (Deposit ABC):* If the completed module is quality-verified and accredited (UMC or SAMC), an official **Academic Credit Record (ACR)** is prepared. Once deposited into the national Academic Bank of Credit, the credits become available for formal credit transfer (Pindahan Kredit) toward a degree at UTeM or any participating higher education institution in Malaysia.
4. **The Single Redemption Principle:** In compliance with MQA governance [Guide Clause 12.9], an academic credit can only be counted *once* toward the award of a formal academic qualification.

Once a credit is redeemed toward an awarded degree, its status transitions permanently to *Redeemed* and it cannot be re-used for a second qualification.

3.4 Accreditation Status & Quality Verification

Offerings on UCreds are rigorously bench-marked against national regulatory standards:

- **MQA Provisional & Full Accreditation:** Flagship postgraduate degrees offered via Open and Distance Learning carry formal accreditation. For instance, the *Master of Technology in Data Science and Analytics* bears accreditation reference number **MQA/PA17473**.
- **APEL Integration (Prior Experiential Learning):** In accordance with national lifelong learning mandates, admission into UCreds ODL programmes supports **APEL.A** (Accreditation of Prior Experiential Learning for Access) for candidates lacking formal academic prerequisites, as well as **APEL.C** (Credit Award) allowing work experience to substitute for specific modular requirements.

4. Comprehensive Course Directory (The 60 Public Offerings)

4.1 Public Catalog Overview & Statistical Breakdown

The public UCreds catalog comprises exactly **60 curated offerings** distributed across three functional types: **ODL Academic Degree Programmes** (3 offerings), **Course Learning Paths** (2 offerings), and **Specialised Micro-Courses** (55 offerings). The overall pricing structure ranges from completely free open-access modules to RM 150.00 for technical certifications, and up to RM 25,110.00 for full postgraduate degree tuition.

Domain Classification	Course Count	Typical Duration	Fee Range (MYR)	Target Audience
ODL Master Degree Programmes	3	1.5 - 2.0 Years	RM 22,610 - 25,110	Postgraduate Scholars & Executives
Structured Learning Paths	2	30 - 60 Hours	Free - RM 150.00	Data Engineers & AI Practitioners
Mathematics & Computational Science	12	10 - 18 Hours	RM 12.00 - Free	Engineering & Computer Science Students
AI, Machine Learning & Analytics	10	12 - 25 Hours	RM 12.00 - Free	Industry Analysts & Software Engineers
Cybersecurity & Digital Hygiene	5	8 - 15 Hours	RM 12.00 - Free	Corporate Employees & Public Learners
Database Systems & Big Data	4	10 - 18 Hours	RM 12.00 - Free	Database Administrators & Developers
Programming Languages & Dev	8	10 - 30 Hours	RM 12.00 - Free	Novice Programmers & Mobile Developers
TVET, Robotics & Engineering	7	8 - 20 Hours	RM 12.00 - Free	TVET Educators & Mechanical Engineers
Languages, Soft Skills & Humanities	7	8 - 15 Hours	RM 12.00 - Free	Working Professionals & General Public
Academic Research & Methodology	2	10 - 15 Hours	Free	Postgraduate & Final Year Researchers

Table 2: *Statistical Distribution of the 60 Public UCreds Offerings*

4.2 ODL Master Degree Programmes

These fully accredited postgraduate programmes are delivered in Open and Distance Learning mode, combining asynchronous video lectures, modular assessments, and remote dissertation supervisions.

ID	Programme Title	Level	Duration	Total Fee	Accreditation Ref.
78	Master of Business Administration	Master	1600 Hours 00 Minutes	MYR 22,610.00	Universiti Teknikal Malaysia Mel...
80	Master of Technology in Data Science and Analytics	Master	1600 Hours 00 Minutes	MYR 25,110.00	Universiti Teknikal Malaysia Mel...
1560	Master of Electronic Engineering (Computer Engineering)	Master	1600 Hours 00 Minutes	-	Universiti Teknikal Malaysia Mel...

Table 3: *Accredited Postgraduate Programmes Delivered via UCreds*

4.3 Mathematics, Calculus & Computational Methods

A flagship series authored by the Centre for Mathematical Sciences and Faculty of Electrical Engineering, covering advanced calculus, transform methods, and foundation numeracy.

ID	Course Title	Level	Duration	Fee	Author / Faculty
155	First Order Differential Equations	Intermediate	18 Hours 00 Minutes	MYR 12.00	Khoo Chin Foon | Loh Ser Le...
140	Second Order Linear Differential Equations	Intermediate	18 Hours 00 Minutes	MYR 12.00	Khoo Chin Foon | Loh Ser Le...
138	Laplace Transform	Intermediate	33 Hours 00 Minutes	MYR 12.00	Khoo Chin Foon | Loh Ser Le...
157	Fourier Series	Intermediate	18 Hours 00 Minutes	MYR 10.00	Khoo Chin Foon | Loh Ser Le...
154	Partial Differential Equations	Intermediate	33 Hours 00 Minutes	MYR 12.00	Khoo Chin Foon | Loh Ser Le...
1430	Math is Fun Aras 1: Penambahan dan Penolakan (Lingkungan 0-100, Tahun 1)	Beginner	-	Free	Universiti Teknikal Malaysia Mel...
1431	Math is Fun Aras 2: Penambahan dan Penolakan (Lingkungan 1000, Tahun 2)	Beginner	-	Free	Universiti Teknikal Malaysia Mel...
1432	Math is Fun Aras 3: Penambahan, Penolakan, Darab, dan Bahagi (Lingkungan 10,000, Tahun 3)	Beginner	-	Free	Universiti Teknikal Malaysia Mel...

ID	Course Title	Level	Duration	Fee	Author / Faculty
1433	Math is Fun Aras 4: Operasi Aritmetik Lanjutan (Lingkungan 100,000, Tahun 4)	Beginner	-	Free	Universiti Teknikal Malaysia Mel...
1434	Math is Fun Aras 5: Operasi Aritmetik Kompleks (Lingkungan 1,000,000, Tahun 5)	Beginner	-	Free	Universiti Teknikal Malaysia Mel...
1435	Math is Fun Aras 6: Penguatan Operasi Aritmetik (Lingkungan 10,000,000, Tahun 6)	Beginner	-	Free	Universiti Teknikal Malaysia Mel...
661	Application of Graph Theory: Network Modelling	Intermediate	11 Hours 00 Minutes	MYR 29.90	Dr Norhazwani Md Yunos | Un...
1425	MATH is FUN Juara Asas Pengiraan	Beginner	-	Free	Universiti Teknikal Malaysia Mel...

Table 4: *Mathematics and Calculus Micro-Credential Offerings*

4.4 Artificial Intelligence, Data Science & Machine Learning

Courses designed to build proficiency in modern machine learning architectures, statistical modelling, and big data visualization, mapped directly to the MTDL curriculum.

ID	Course Title	Level	Duration	Fee	Author / Faculty
515	Embracing Machine Learning for Business Decision [PM Choo]	Beginner	06 Hours 30 Minutes	MYR 50.00	Universiti Teknikal Malaysia Mel...
517	7. [MAXL5143] Introduction to Deep Learning	Intermediate	07 Hours 30 Minutes	Free	Dr. Noor Fazilla Abd Yusof |...
530	6.[MAXL5113] Overview of Analytics Models	Beginner	06 Hours 00 Minutes	MYR 50.00	Zeratul Izzah | Universiti ...
435	Data Modeling	Beginner	16 Hours 00 Minutes	MYR 84.00	Universiti Teknikal Malaysia Mel...
262	Artificial Intelligence for Everyone	Beginner	08 Hours 00 Minutes	MYR 10.00	Syafeeza Ahmad Radzi | Univ...
519	Time Series Forecasting	Intermediate	08 Hours 00 Minutes	MYR 150.00	Dr. Siti Azirah Asmai | Uni...

ID	Course Title	Level	Duration	Fee	Author / Faculty
665	Kick-start Your Ambition as Machine Learning Engineer	Beginner	08 Hours 00 Minutes	MYR 9.99	Dr. Noor Fazilla Abd Yusof |...
817	Working with Machine Learning by Examples	Intermediate	15 Hours 00 Minutes	MYR 19.99	Assoc. Prof. Dr. Choo Yun Huoy &...
1296	MAXL5273 Customer and Financial Analytics [Elective]	Intermediate	120 Hours 00 Minutes	Free	Universiti Teknikal Malaysia Mel...
648	Journey Towards Artificial Intelligence	Beginner	10 Hours 00 Minutes	MYR 10.00	Dr. Nur Zareen Zulkarnain |...

Table 5: *Artificial Intelligence and Data Science Micro-Courses*

4.5 Cybersecurity, Information Security & Privacy

Focusing on proactive threat awareness, phishing countermeasures, home network security, and credential management for both corporate staff and everyday citizens.

ID	Course Title	Level	Duration	Fee	Author / Faculty
486	Phishing Techniques	Beginner	04 Hours 00 Minutes	MYR 9.99	Dr. Nur Fadzilah Othman | U...
436	Social Engineering	Beginner	04 Hours 00 Minutes	MYR 9.99	Dr. Nur Fadzilah Othman | U...
487	Security at home	Beginner	04 Hours 00 Minutes	MYR 9.99	Ts. Erman Hamid | Universit...
489	Password and Authentication 101	Beginner	04 Hours 00 Minutes	MYR 9.99	Ts. Dr. Mohd Fairuz Iskandar Oth...
543	Public Wi-Fi Security	Beginner	04 Hours 00 Minutes	MYR 9.99	Ts. Dr. Mohd Fairuz Iskandar Oth...

Table 6: *Cybersecurity and Information Hygiene Courses*

4.6 Database Engineering & Big Data Management

Structured training covering relational database design, relational normalization theory, NoSQL architectures, and distributed storage.

ID	Course Title	Level	Duration	Fee	Author / Faculty
520	1. [MAXL5123] Big Data Management Demystified	Intermediate	07 Hours 00 Minutes	MYR 50.00	Universiti Teknikal Malaysia Mel...
644	Fundamentals of a Database	Beginner	16 Hours 00 Minutes	MYR 84.00	Noor Azilah binti Draman@Muda &#...
512	Maximize Your Database Efficiency: A Normalization Practical Guide	Beginner	16 Hours 00 Minutes	MYR 80.00	Universiti Teknikal Malaysia Mel...

Table 7: *Database Systems and Storage Management Courses*

4.7 Software Engineering, Programming & Mobile Apps

Practical coding courses ranging from introductory Python and object-oriented Java to cross-platform mobile application development with Flutter.

ID	Course Title	Level	Duration	Fee	Author / Faculty
1883	Python Essentials 1: Module 3	Beginner	-	Free	Universiti Teknikal Malaysia Mel...
2009	Python Essentials 1: Module 4	Beginner	-	Free	SAIL Research Group, Universiti ...
290	Java Programming for Beginners	Beginner	04 Hours 00 Minutes	MYR 50.00	Emaliana Kasmuri | Universi...
571	Easy-Peasy Python Programming	Beginner	08 Hours 00 Minutes	MYR 20.00	Universiti Teknikal Malaysia Mel...
827	Membina Aplikasi Flutter: Panduan Praktikal	Beginner	25 Hours 00 Minutes	MYR 150.00	Universiti Teknikal Malaysia Mel...
1541	PlaCe-GMC: Player-Based JavaScript Journey	Beginner	08 Hours 00 Minutes	Free	Sazilah Salam | Zuhri Arafa...
1881	Level 1: Hello, World! — Basics of Python Programming	Beginner	120 Hours 00 Minutes	Free	Universiti Teknikal Malaysia Mel...

Table 8: *Software Development and Programming Language Offerings*

4.8 Advanced TVET, Robotics & Mechanical Engineering

Reflecting UTeM's mission as a premier Technical University (MTUN), these modules cover mobile robotics, automated control charts, 3D printing workflows, and medical CAD modeling.

ID	Course Title	Level	Duration	Fee	Author / Faculty
132	Industrial Robotics for Industry 4.0	Beginner	08 Hours 00 Minutes	MYR 5.00	Mohd Nazmin Maslan | Univer...
309	3D Design using AUTODESK INVENTOR as easy 1 2 3	Beginner	01 Hours 25 Minutes	MYR 10.00	Dr. Ihwan Ghazali, Ts. Khairul A...
498	Getting Started with STM32 IoT	Beginner	06 Hours 00 Minutes	MYR 19.00	Sani Irwan Salim | Universi...
499	Buoyancy: Float the body, or sink it?	Beginner	08 Hours 00 Minutes	MYR 2.00	Dr. Ernie Mat Tokit | Unive...
568	Control Chart for Beginners in TVET Education	Beginner	08 Hours 00 Minutes	MYR 39.90	Yusliza binti Yusuf | Unive...
573	Mudahnya Robot Mudah Alih!	Beginner	08 Hours 00 Minutes	MYR 15.00	Ahmad Zaki bin Shukor | Nur...
580	CAD Modeling: Easy Modeling Vertebrae Bone using MRI Images	Intermediate	08 Hours 00 Minutes	MYR 10.00	Masni-Azian Akiah | Univers...
659	3D Scanning to 3D Printing	Beginner	08 Hours 00 Minutes	MYR 20.00	Dr. Mohd Najib Ali Mokhtar, Univ...
808	Visualise Product Through Packaging	Beginner	08 Hours 00 Minutes	MYR 10.00	Rosidah Jaafar, Mohd Azlan bin M...
809	Teknologi dan Kejuruteraan Sistem Kawalan dalam Kehidupan	Beginner	08 Hours 00 Minutes	MYR 20.00	Universiti Teknikal Malaysia Mel...

Table 9: *Technical, Vocational & Engineering (TVET) Micro-Credentials*

4.9 Languages, Communication & Soft Skills

Interdisciplinary courses designed to foster global communication, cultural fluency (Mandarin phonetics and character stroke order), and executive public speaking.

ID	Course Title	Level	Duration	Fee	Author / Faculty
412	The Essentials of Critical and Creative Thinking for Innovation	Beginner	04 Hours 00 Minutes	MYR 4.99	Dr. Zanariah Jano | Univers...
182	Speak Mandarin Using Chinese Phonetics	Beginner	12 Hours 00 Minutes	Free	Dr. Cheong Kar Mee | Univer...
184	The Wonders of Chinese Characters	Beginner	12 Hours 00 Minutes	Free	Tan Poh Ee | Universiti Tek...
553	Berani Berucap	Beginner	42 Hours 00 Minutes	MYR 39.00	Universiti Teknikal Malaysia Mel...
570	Beautiful Photo with Smartphone Photography Techniques	Beginner	04 Hours 00 Minutes	MYR 10.00	Atiqah Deni | Zulisman Maks...
1776	MPU21072 Pengenalan Penghayatan Etika dan Peradaban Dalam Acuan Malaysia	Beginner	120 Hours 00 Minutes	Free	Politeknik Metro Johor Baharu &#...
1513	Writing Literature Review	Beginner	07 Hours 00 Minutes	MYR 100.00	Assoc. Prof. Dr. Nurul Akmar Emr...
1770	Literature Search (FTMK)	Beginner	06 Hours 00 Minutes	MYR 100.00	Assoc. Prof. Dr. Nurul Akmar Emr...

Table 10: *Languages, Humanities, and Academic Research Modules*

4.10 The Underlying Moodle Repository (214 Course Instances)

While the public catalog exposes 60 polished entry-points, direct inspection of the underlying Moodle core directory (`/course/index.php`) reveals **214 registered course instances**. This deeper repository contains internal subject modules (such as the full `MAXL` and `MECL` series), faculty staging sandboxes, and modular assessment units that feed into the public micro-credentials.

5. Flagship Course Profiles & In-Depth Syllabi

5.1 Profile 1: Master of Technology in Data Science and Analytics (ODL) [ID 80]

Parameter	Specification / Details
Programme Code	MTDL (Open and Distance Learning)
Accreditation Ref.	**MQA/PA17473** (Malaysian Qualifications Agency Registered)
Total Tuition Fee	**MYR 25,110.00** (Local Learner Schedule)
Host Faculty	Faculty of Information and Communication Technology (FTMK)
Duration / Mode	1.5 to 3 Years | 100% Online Asynchronous with Synchronous Mentoring
Target Audience	Data Analysts, IT Executives, Statisticians, and Software Engineers seeking advanced analytics expertise.
Entry Requirements	Bachelor's Degree in Computing/Science/Technology (min CGPA 2.50) OR Bachelor's Degree with working experience OR **APEL.A (T-7)** certification.
Core Curriculum	MAXL5113 Fundamental of Data Science, MAXL5123 Big Data Management, MAXL5133 Applied Statistical Methods, MAXL5143 Applied Machine Learning, MAXL5153 Big Data Analytics and Visualization, MAXL5163 Modelling and Decision Making, MPSW5013 Research Methodology, MAXL5223 Group Project.
Programme Learning Outcomes (PLOs)	• Synthesize advanced knowledge in data science principles, machine learning models, and big data architectures to solve industry problems. • Design, deploy, and evaluate scalable data pipelines utilizing Apache Spark, Hadoop, and modern NoSQL datastores. • Demonstrate professional ethics, data privacy compliance, and effective communication of analytical findings to executive stakeholders.

5.2 Profile 2: Master of Business Administration (ODL) [ID 78]

Parameter	Specification / Details
Programme Code	MBA (Open and Distance Learning Mode)
Total Tuition Fee	**MYR 22,610.00**
Host Faculty	Faculty of Technology Management and Technopreneurship (FPTT)
Duration / Mode	1.5 Years (Trimester Structure) | 100% Online Distance Learning
Entry Requirements	Recognised Bachelor's degree with minimum CGPA 2.50 OR relevant industrial managerial experience OR **APEL.A Level 7**.
Core Pillars	Strategic Management, Managerial Economics, Technopreneurship, Corporate Finance, Human Capital Analytics, and Business Research Dissertation.
Programme Learning Outcomes	• Integrate strategic management theory with technopreneurial execution in competitive global markets. • Lead organizational transformation through data-informed financial and operational decision-making. • Formulate ethical corporate governance and corporate social responsibility (CSR) policies.

5.3 Profile 3: First Order Differential Equations [ID 155] (Mathematics Series)

Parameter	Specification / Details

Course Level	Intermediate | Micro-Course
Student Learning Time	**18 Hours 00 Minutes**
Enrolment Fee	**MYR 12.00**
Instructors	Khoo Chin Foon, Loh Ser Lee, Irma Wani Binti Jamaludin, Rahifa Binti Ranom, Suhaila Binti Salleh (Centre for Mathematical Sciences)
Prerequisites	Basic Single-Variable Differential and Integral Calculus.
Syllabus Breakdown	Part 1: Introduction and classification of ordinary differential equations (order and linearity). Part 2: Exact differential equations and integrating factors. Part 3: Separable equations, homogeneous substitutions, and linear first-order differential equations with real-world engineering boundary conditions.
Micro Learning Outcomes (MLOs)	• Classify differential equations according to order, degree, and linearity. • Determine whether a given first-order differential equation is exact and apply integrating factors when inexact. • Solve initial value problems (IVPs) modeling physical electrical circuits (RL/RC) and thermal cooling systems.

5.4 Profile 4: Introduction to Deep Learning (MAXL5143) [ID 517]

Parameter	Specification / Details
Course Level	Intermediate / Modular Postgraduate Micro-Credential
Student Learning Time	15 Hours SLT
Enrolment Fee	**MYR 12.00**
Lead Instructor	Dr. Noor Fazilla Abd Yusof (Faculty of Information and Communication Technology)
Prerequisites	Foundations of Python Programming and Linear Algebra.
Syllabus Breakdown	Biological versus artificial neurons, multi-layer perceptron (MLP) architectures, forward and backpropagation mathematics, loss functions (cross-entropy, MSE), optimization techniques (SGD, Adam), and regularization (Dropout, Batch Normalization).
Micro Learning Outcomes	• Explain the architectural differences between traditional machine learning algorithms and deep neural networks. • Implement a multi-layer deep neural network using modern Python frameworks (TensorFlow / Keras). • Tune hyper-parameters to mitigate overfitting on complex high-dimensional datasets.

5.5 Profile 5: Fundamentals of a Database [ID 644]

Parameter	Specification / Details
Course Level	Beginner / Foundation
Student Learning Time	12 Hours SLT
Enrolment Fee	**MYR 12.00**
Instructors	Noor Azilah binti Draman@Muda, Noraswaliza binti Abdullah (FTMK)
Prerequisites	None (Introductory Level).
Syllabus Breakdown	Introduction to database management systems (DBMS), relational database architecture, relational algebra, entity-relationship diagrams (ERD), primary/foreign key constraints, and introductory SQL queries (SELECT, JOIN, GROUP BY).

Micro Learning Outcomes	• Differentiate between flat file systems and relational database management systems. • Construct Entity-Relationship Diagrams (ERD) that accurately represent business rules. • Execute standard SQL Data Definition (DDL) and Data Manipulation (DML) statements.

5.6 Profile 6: Mudahnya Robot Mudah Alih! [ID 573] (Mobile Robotics)

Parameter	Specification / Details
Course Level	Beginner / TVET Focus
Student Learning Time	10 Hours SLT
Enrolment Fee	**MYR 12.00**
Instructors	Faculty of Electrical Engineering (FKE) Robotics Research Group
Prerequisites	Basic interest in electronics and microcontroller logic.
Syllabus Breakdown	Chassis assembly, DC motor selection and H-bridge motor drivers (L298N), ultrasonic distance sensor interfacing, infrared line-following algorithms, and microcontroller integration (Arduino / ESP32).
Micro Learning Outcomes	• Identify fundamental hardware components of differential-drive mobile wheeled robots. • Assemble and interface sensory hardware with microcontrollers for obstacle avoidance. • Program autonomous line-tracking navigation routines in C/C++.

5.7 Profile 7: Phishing Techniques & Cyber Hygiene [ID 486]

Parameter	Specification / Details
Course Level	Beginner / General Awareness
Student Learning Time	8 Hours SLT
Enrolment Fee	**MYR 12.00**
Instructors	Faculty of Information and Communication Technology (FTMK) Information Security Group
Prerequisites	None (General Public).
Syllabus Breakdown	Anatomy of social engineering attacks, deceptive email header analysis, URL spoofing techniques, malicious attachments (macros, PDF exploits), credential harvesting landing pages, and institutional incident reporting procedures.
Micro Learning Outcomes	• Analyze inbound electronic messages for hallmarks of social engineering and spear phishing. • Verify link integrity and domain certificate validation before submitting credentials. • Execute proactive defense practices against malware delivery in institutional workspaces.

6. Student Support Infrastructure, Knowledge Base & Policies

6.1 Central Helpdesk & Technical Support

To ensure high learner retention and immediate issue resolution across global time zones, CODL operates a multi-tiered support helpdesk directly linked to the UCreds navigation bar:

- **Support Email:** `ucreds.helpdesk@utem.edu.my` (Standard SLA: Response within 24 operational hours; error reports require mandatory screenshot attachments).
- **Direct Telephone Hotline:** `06-229 2272`.
- **Operational Hours:** 8:00 AM – 5:00 PM, Monday through Friday (Malaysia Time, UTC/GMT +8:00).
- **Virtual Companion / Chatbot:** Embedded 24/7 interactive automated guide assisting learners with navigation, login troubleshooting, and course enrolment guidance.

6.2 The Student Support Knowledge Base (36 Articles Analyzed)

Operating under `https://ucreds.utem.edu.my/support/student/`, the WordPress BetterDocs subsystem provides a comprehensive self-service knowledge base divided into three key clusters:

Documentation Cluster	Article Count	Representative Core Topics Covered	Governance Role
eOrientation & Regulations	14 Articles	ODL Student Handbook, Postgraduate Academic Regulations, Examportal Guide, Tuition Fee Schedule, Course Registration Steps	Mandates academic integrity, grading criteria, and graduation procedures.
eCounseling & Mental Wellness	19 Articles	Career Counseling, Crisis Management & Suicide Prevention, Digital Balance & Social Media, Nutrition & Sleep Hygiene, Stress Management	Provides holistic psychological support for adult distance learners.
Online Campus Facilities	3 Articles	Virtual Library Services, Campus VPN Gateway Setup, Student Welfare Assistance (Unit Kebajikan Pelajar)	Delivers off-campus access to licensed journals, books, and financial aid.

Table 11: *Knowledge Base Taxonomy of the UCreds Support Subsystem*

6.3 Key Institutional Policies and Regulations

Audit of the published articles reveals strict adherence to UTeM Senate postgraduate academic regulations:

1. **Academic Regulations & Grading:** ODL postgraduate courses adhere to the standard UTeM grading scheme where Grade B (CGPA 3.00) is the minimum passing grade for credit accumulation toward degree conferment.
2. **Virtual Library Access:** Learners gain authenticated access to UTeM's digital library portal (`https://library.utem.edu.my/`), providing IEEE Xplore, ScienceDirect, and Scopus subscriptions via institutional EZproxy and VPN tunnels.
3. **Campus VPN Security:** Off-campus distance learners are provisioned with secure, encrypted tunnel access to UTeM's internal intranet devices, protecting against man-in-the-middle attacks and session eavesdropping.
4. **Official ODL Handbook:** The complete operational handbook is accessible at: `https://codl.utem.edu.my/02/wp-content/uploads/OPEN-AND-DISTANCE-LEARNING-HANDBOOK-V1-2025.pdf`.

7. Visual Audit & Complete Portal Interface Gallery

7.1 Comprehensive Visual Evidence

During the automated crawling and forensic exploration of the UCreds platform, **26 high-resolution full-page screenshots** were captured. Every single captured interface is documented in this gallery, organized into four operational domains: platform landing, catalog facets, course syllabus overviews, and student support articles. Each figure is rendered with high fidelity to provide an exhaustive visual record of UCreds.

7.1 Platform Landing, Navigation & Public Entry

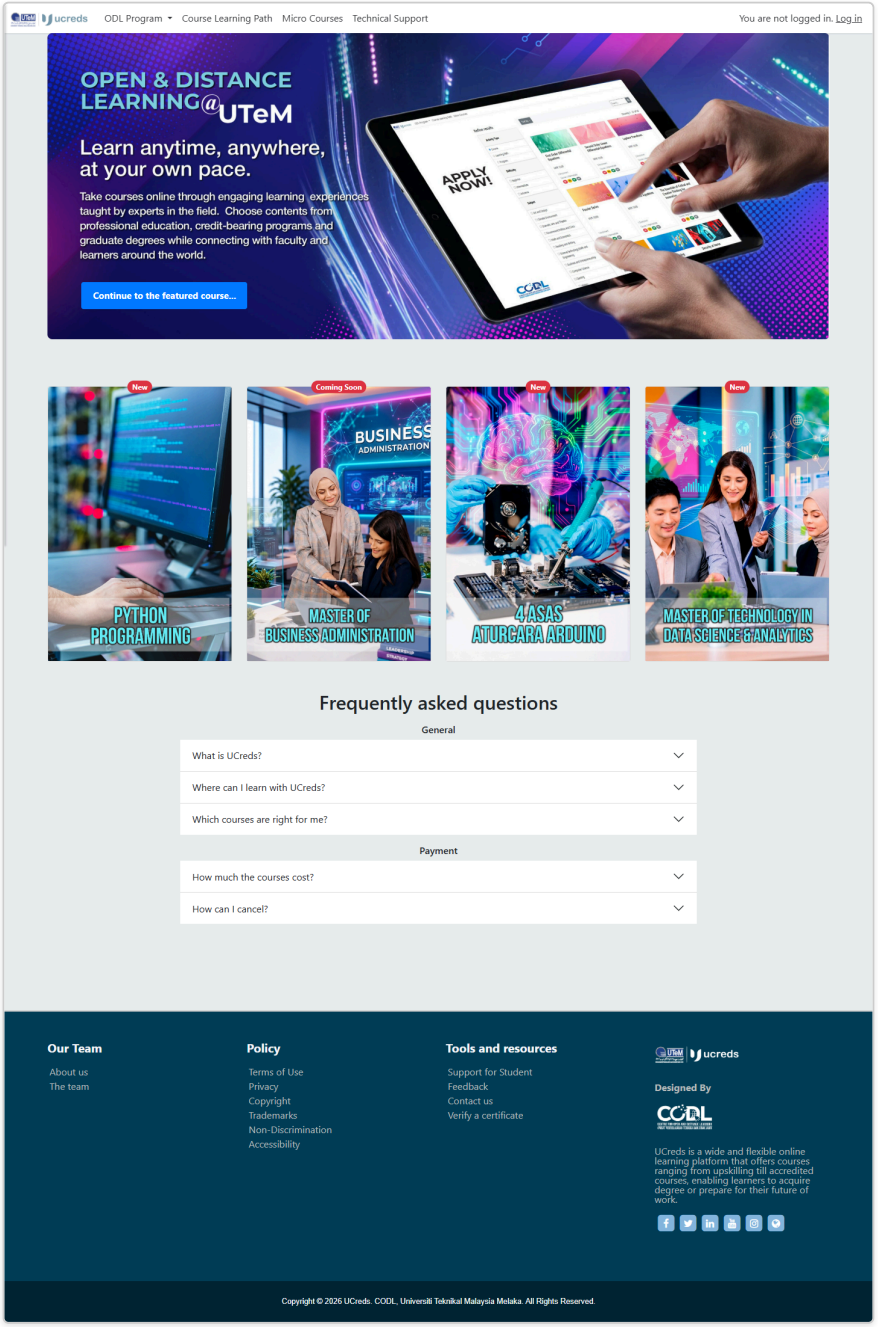

Figure 3: *UCreds Public Portal Homepage featuring Hero Banner, Core Pillars, Value Propositions, and Search Navigation (`assets/01_homepage.png`)*

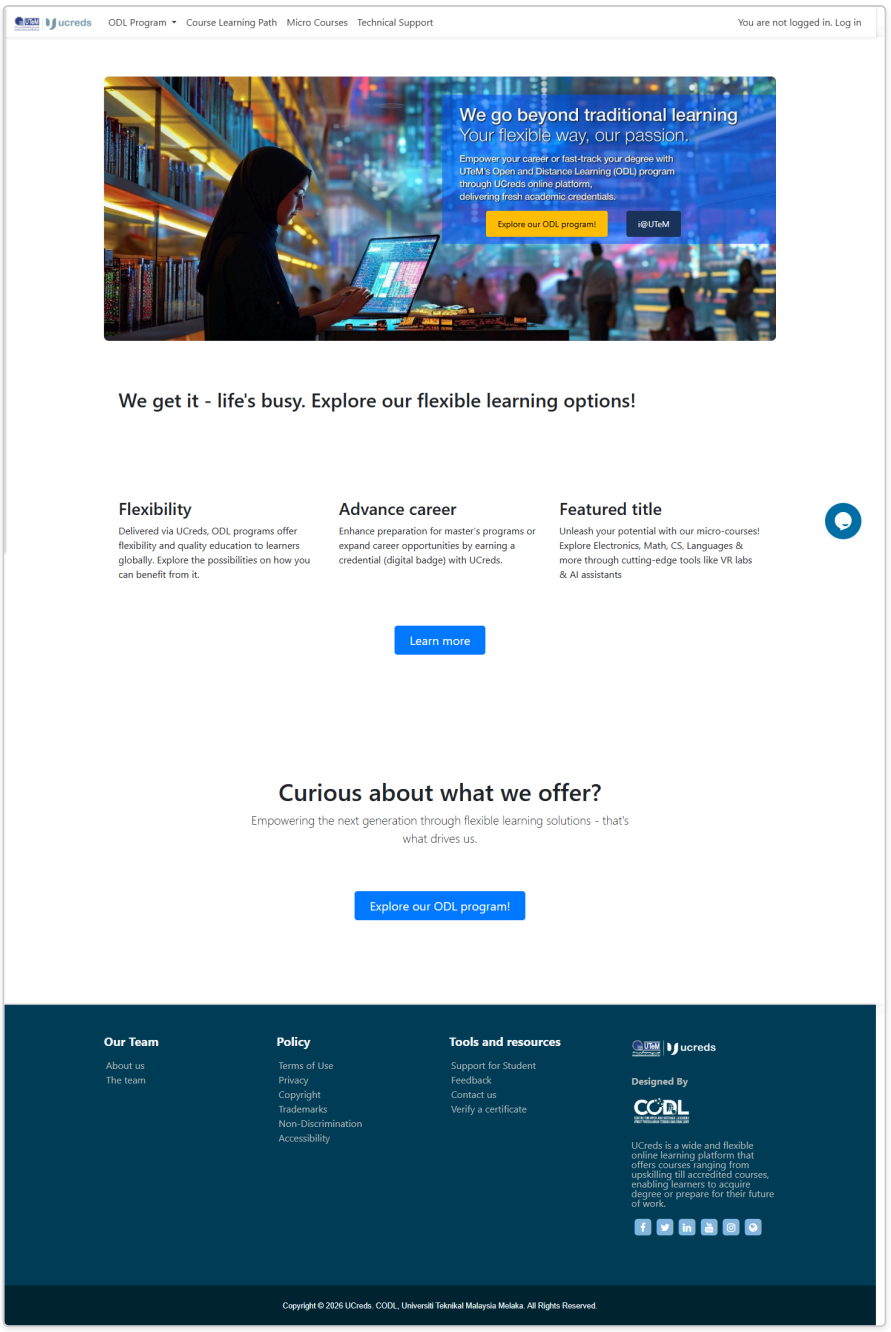

Figure 4: *About ODL Portal Section highlighting Flexible Learning Vision and E-Orientation Onboarding Modal (`assets/07_about_odl.png`)*

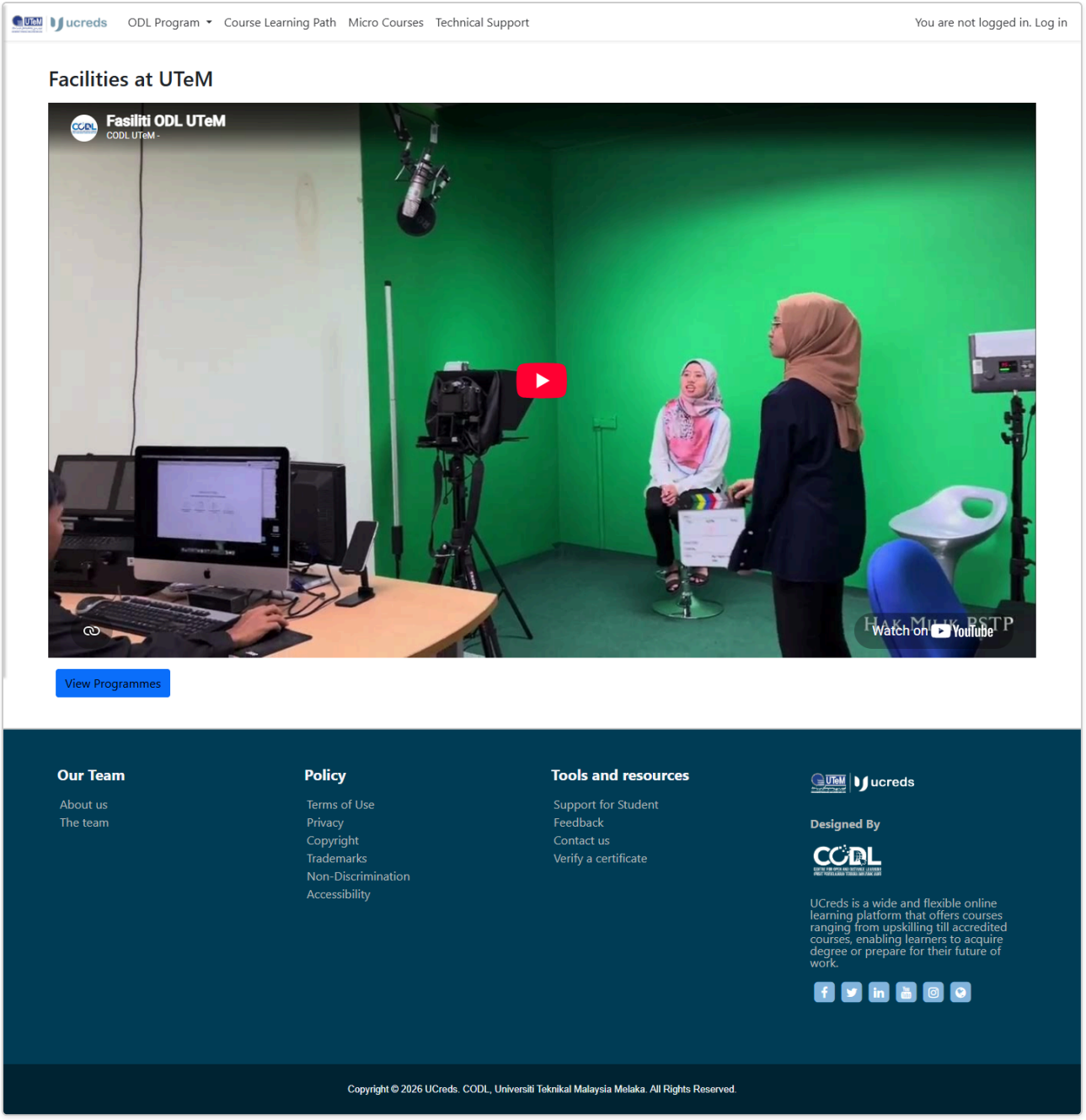

Figure 5: *Facilities at UTeM Section outlining Digital Infrastructure, Campus Laboratories, Virtual Library, and LMS Access (`assets/08_facilities.png`)*

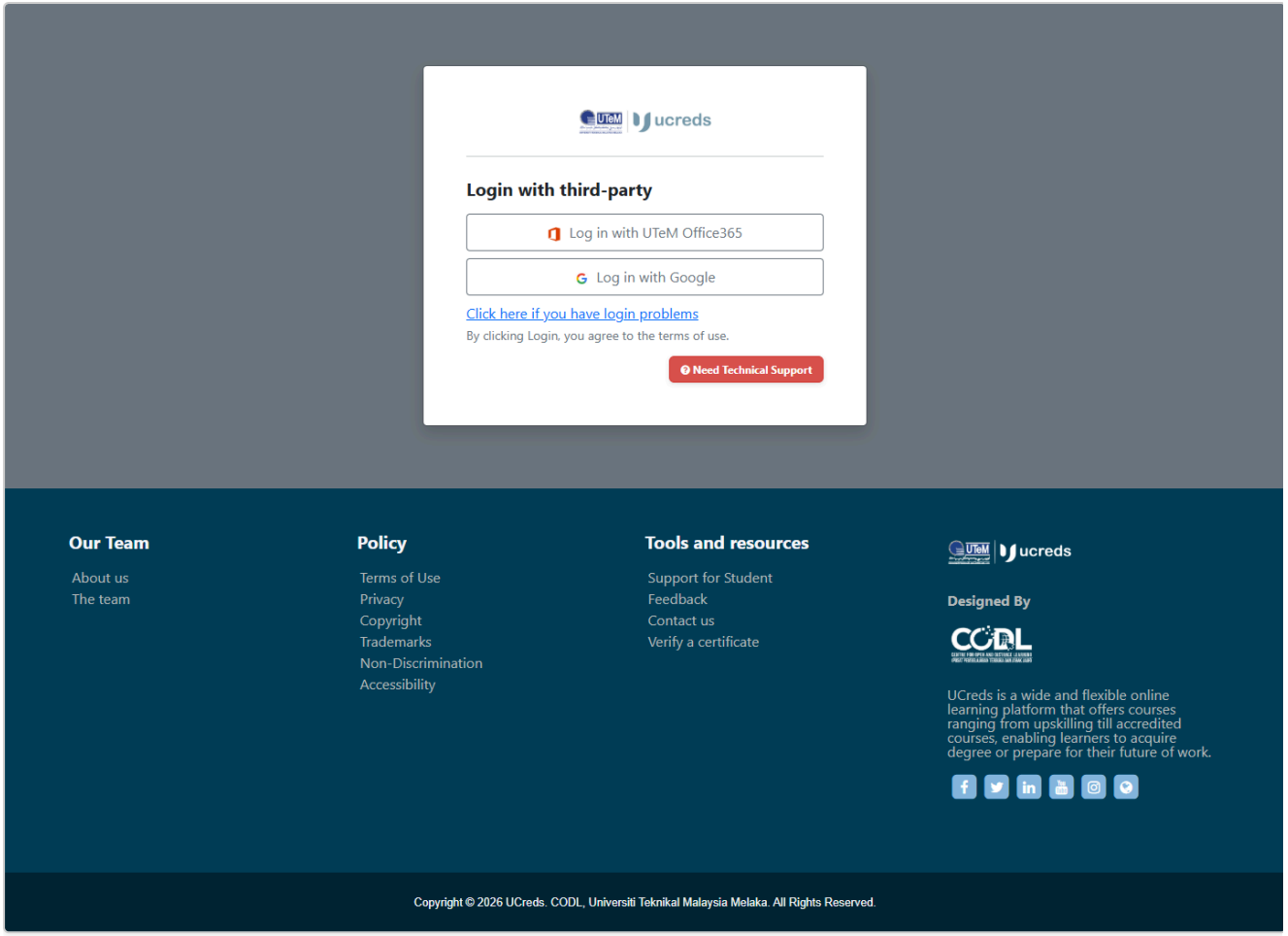

Figure 6: *Unified Authentication Gateway featuring Single Sign-On (Microsoft Entra ID) and Google OAuth2 Integrations (`assets/10_login_portal.png`)*

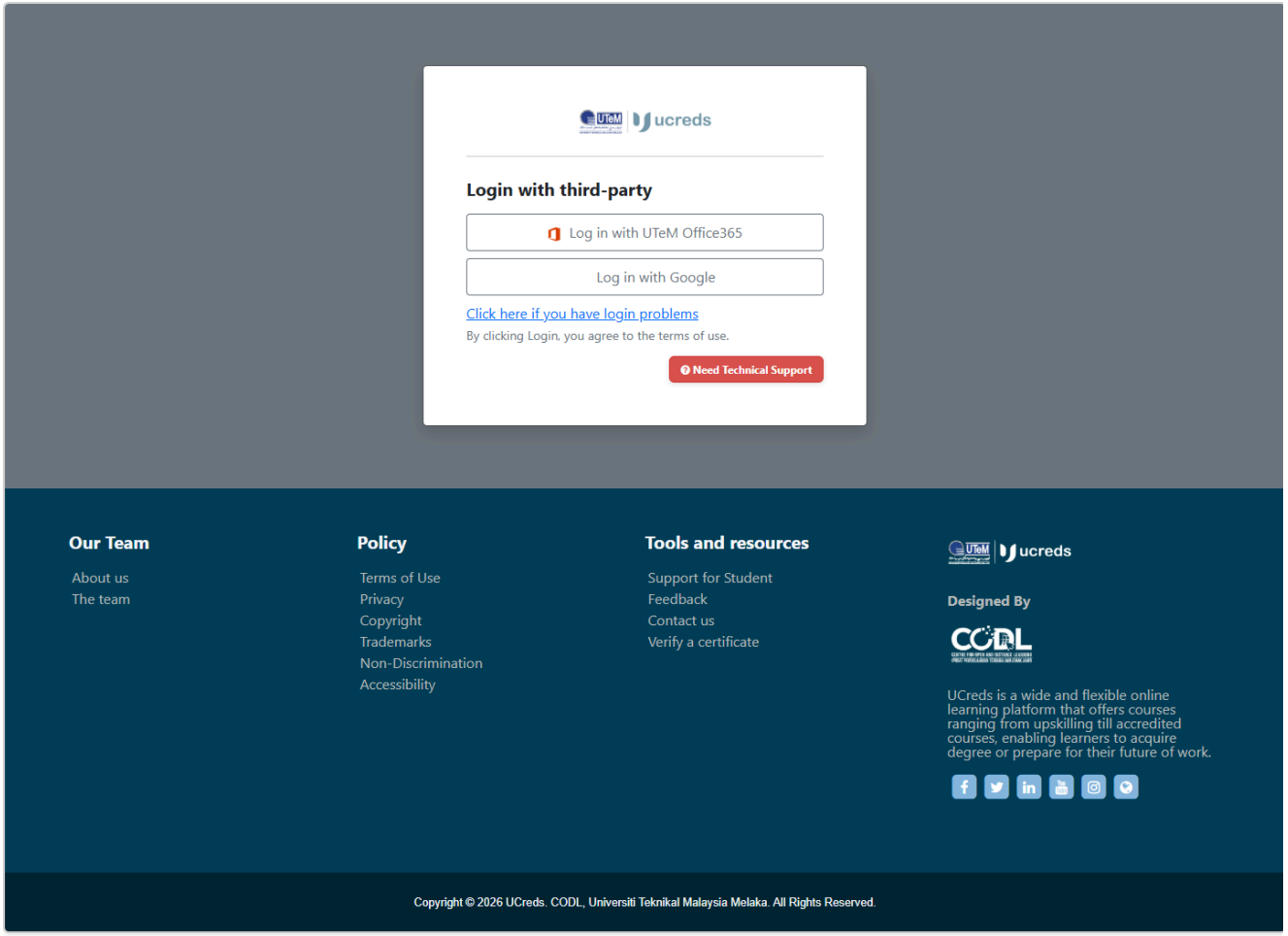

Figure 7: *Native Moodle Internal Login View with Manual Credential Fields and Support Notice (`assets/05_login_page.png`)*

7.2 Catalog Subsystem & Browsing Facets

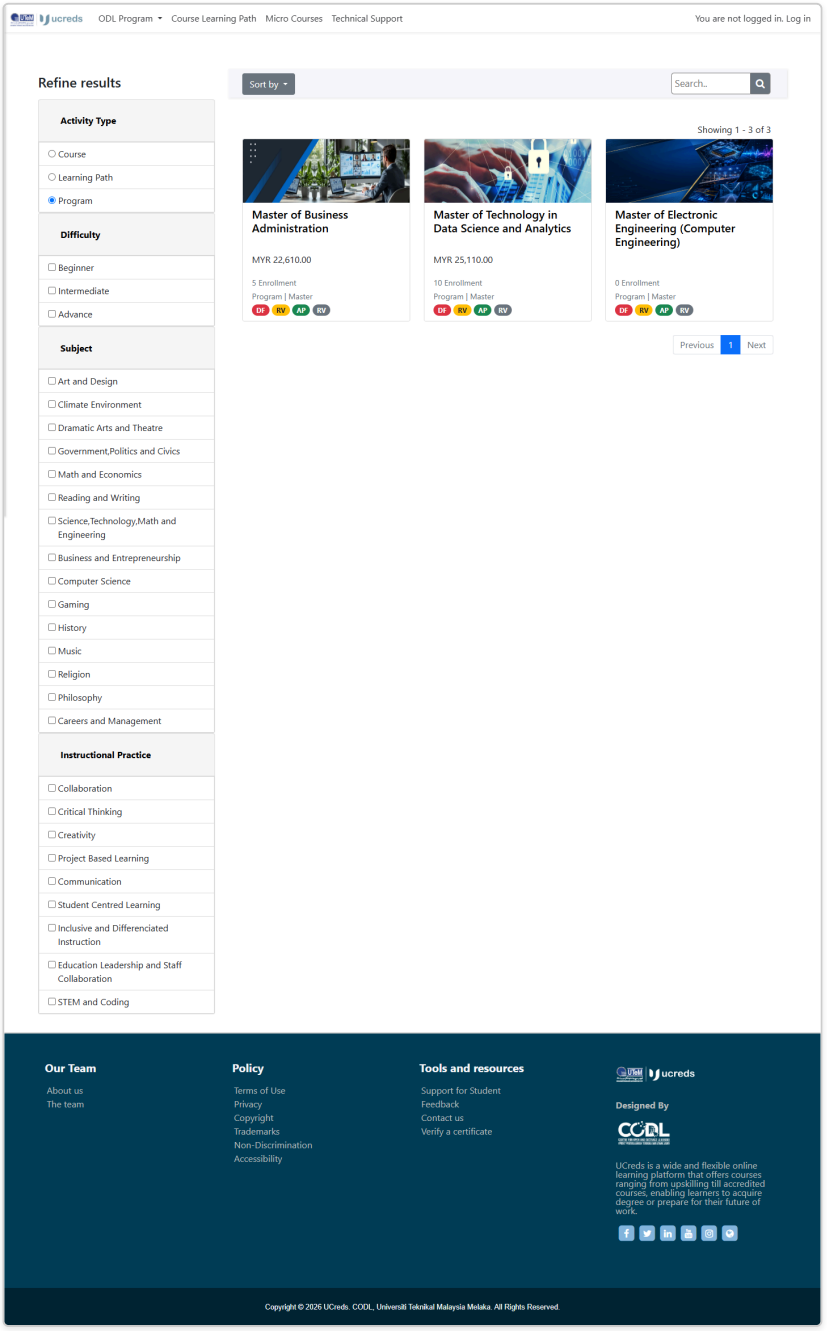

Figure 8: *Postgraduate Degree Catalog displaying MBA and MTDL Open and Distance Learning Offerings (`assets/02_catalog_programmes.png`)*

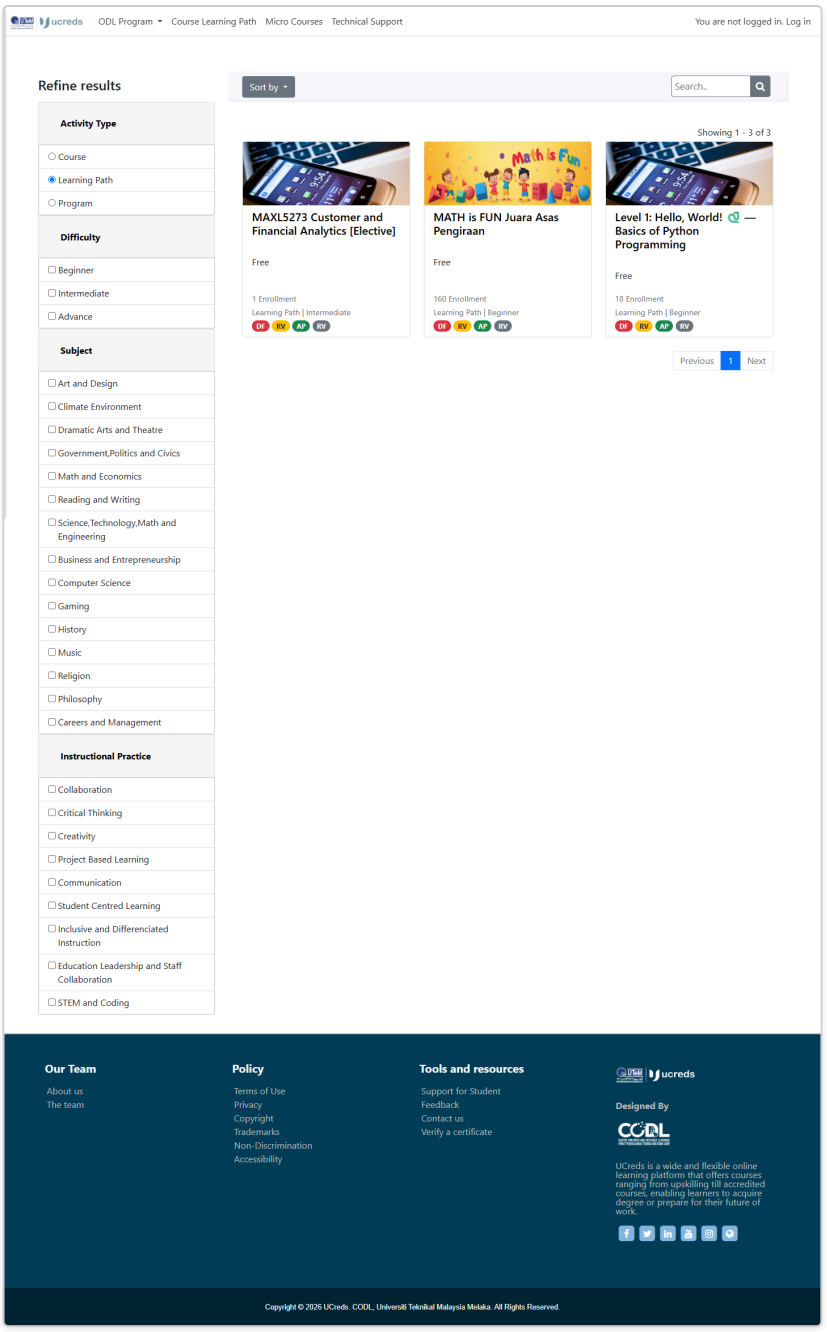

Figure 9: *Catalog Category Filter for Structured Course Learning Paths (`assets/02_catalog_learning_path.png`)*

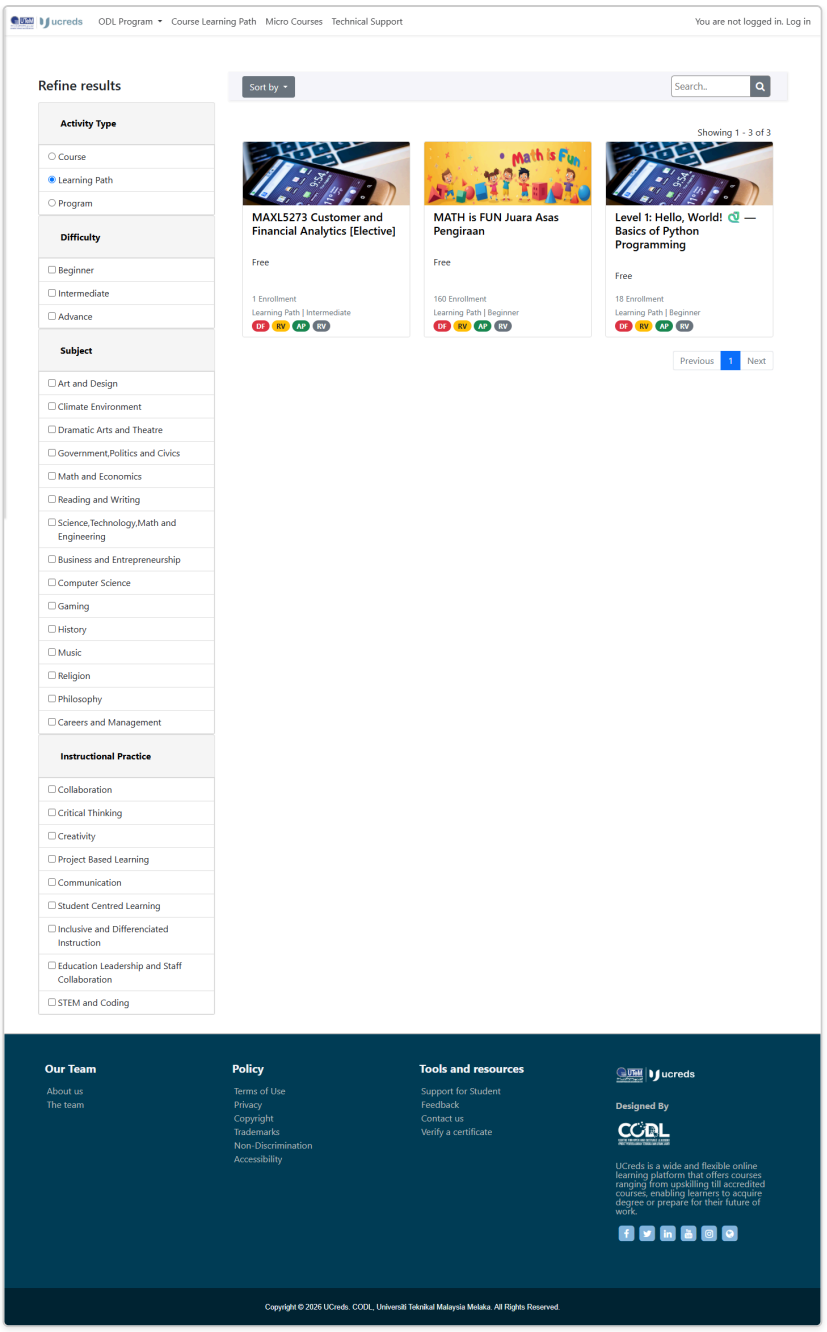

Figure 10: *Course Learning Paths Grid showcasing Specialisation Pathways (`assets/03_course_learning_path.png`)*

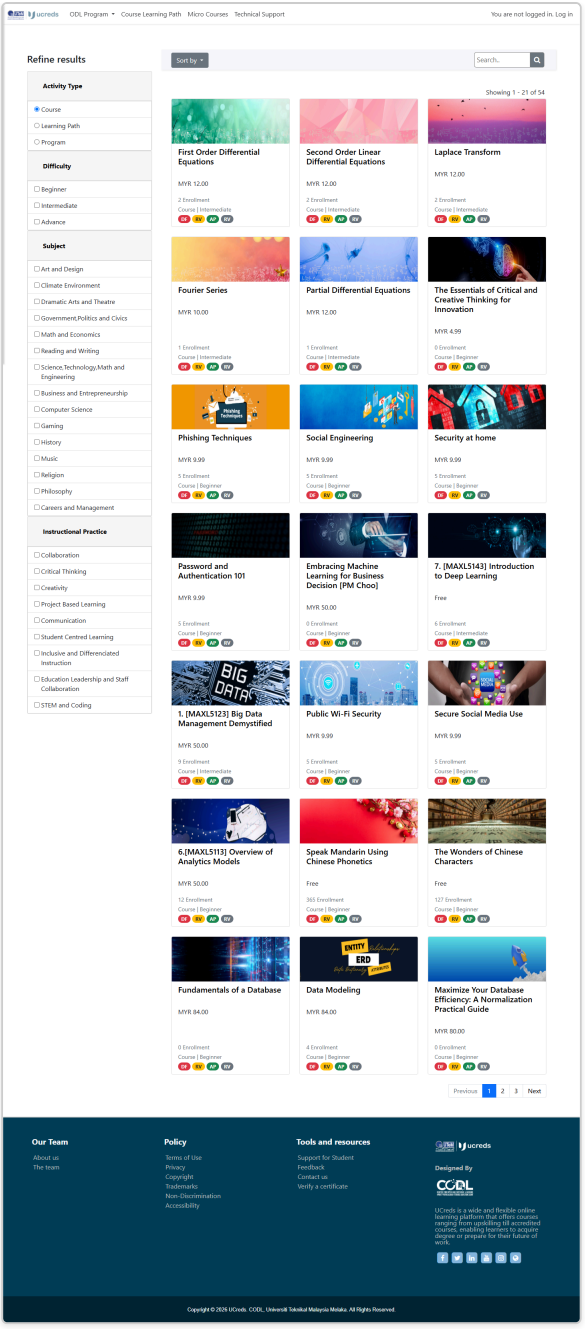

Figure 11: *General Public Catalog Interface with Combined Course Search and Category Facets (`assets/04_catalog_general.png`)*

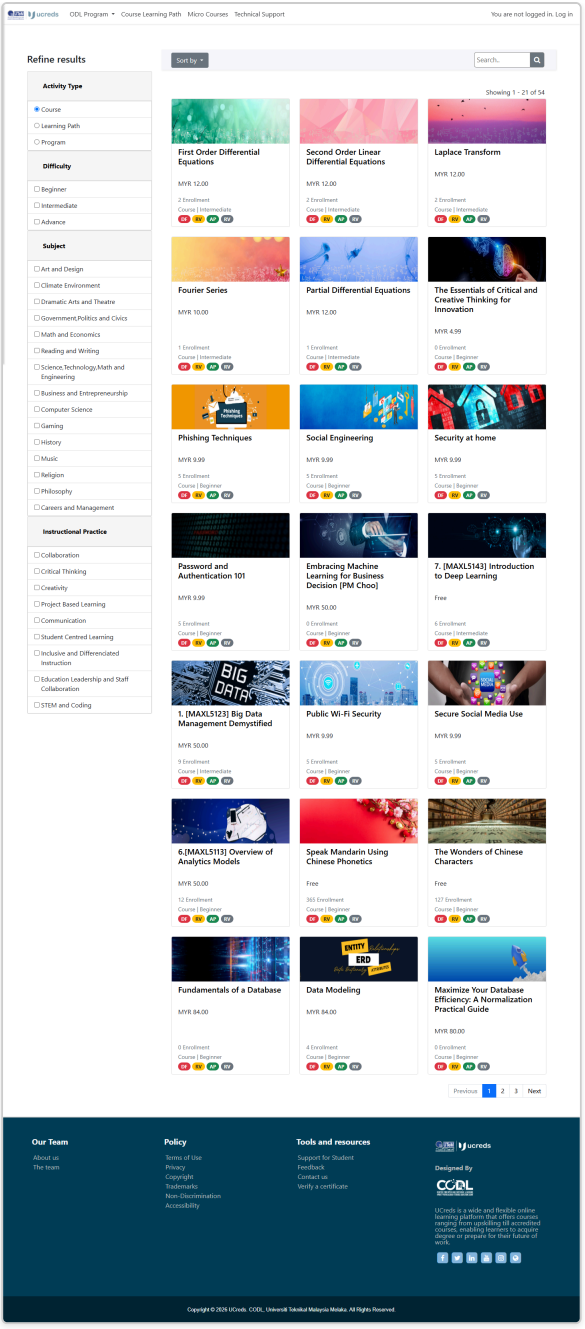

Figure 12: *Catalog Filter View focused on Standalone Micro-Courses (`assets/03_catalog_micro_courses.png`)*

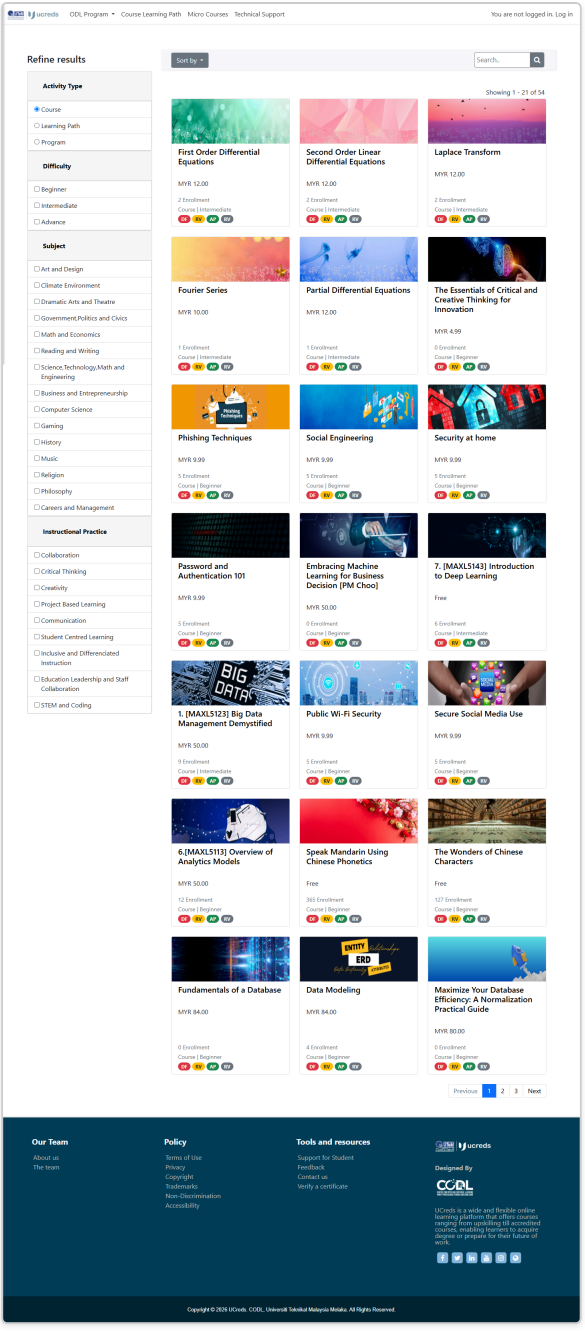

Figure 13: *Micro-Courses Catalog (Page 1) covering Mathematics, AI, Cybersecurity, and Database Foundations (`assets/04_micro_courses_page1.png`)*

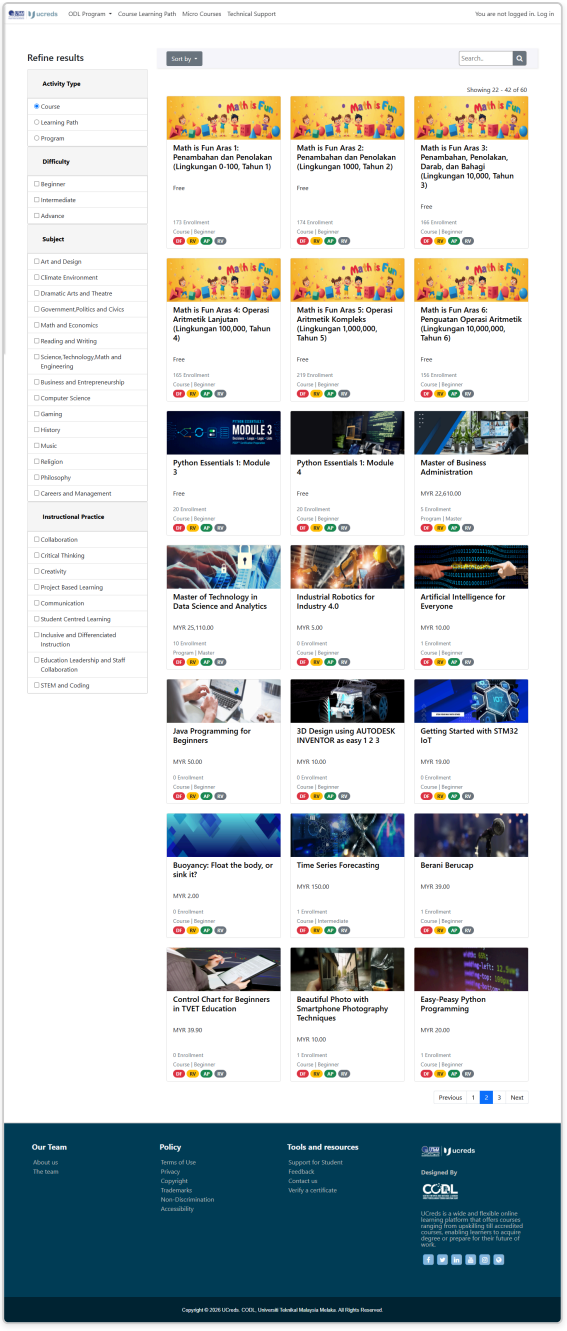

Figure 14: *Micro-Courses Catalog (Page 2) covering Python Programming, Robotics, TVET Education, and CAD Modeling (`assets/05_micro_courses_page2.png`)*

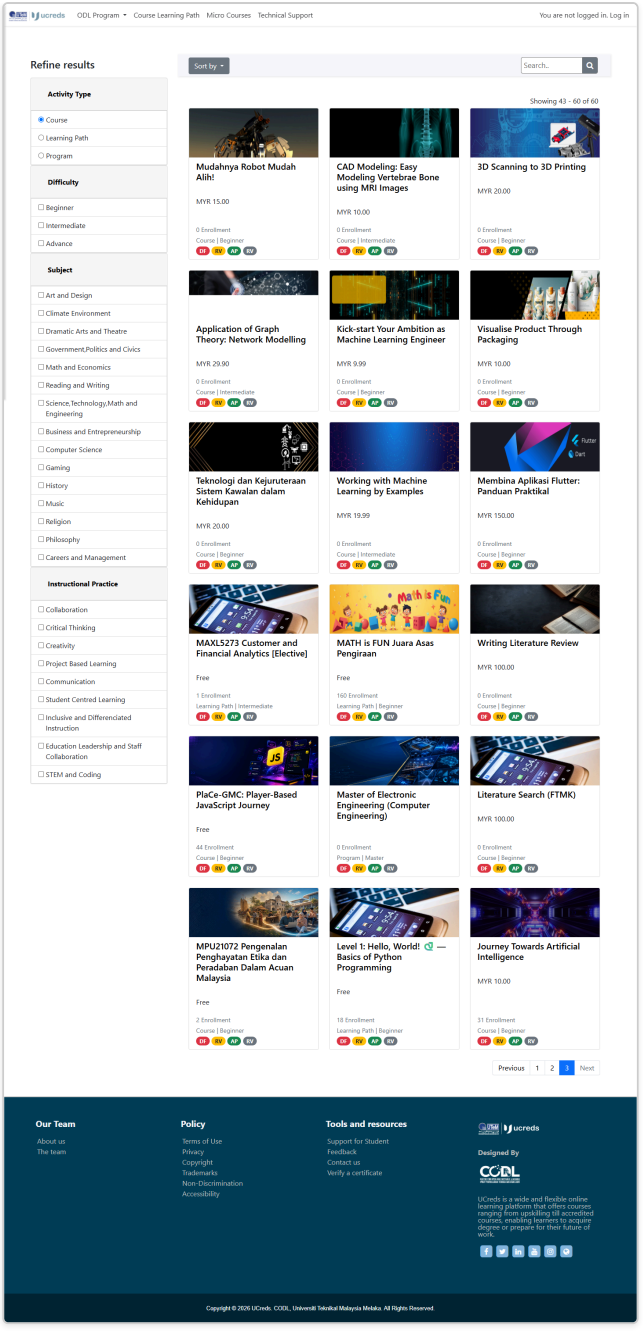

Figure 15: *Micro-Courses Catalog (Page 3) covering Flutter Development, 3D Printing, Mandarin, and Research (`assets/06_micro_courses_page3.png`)*

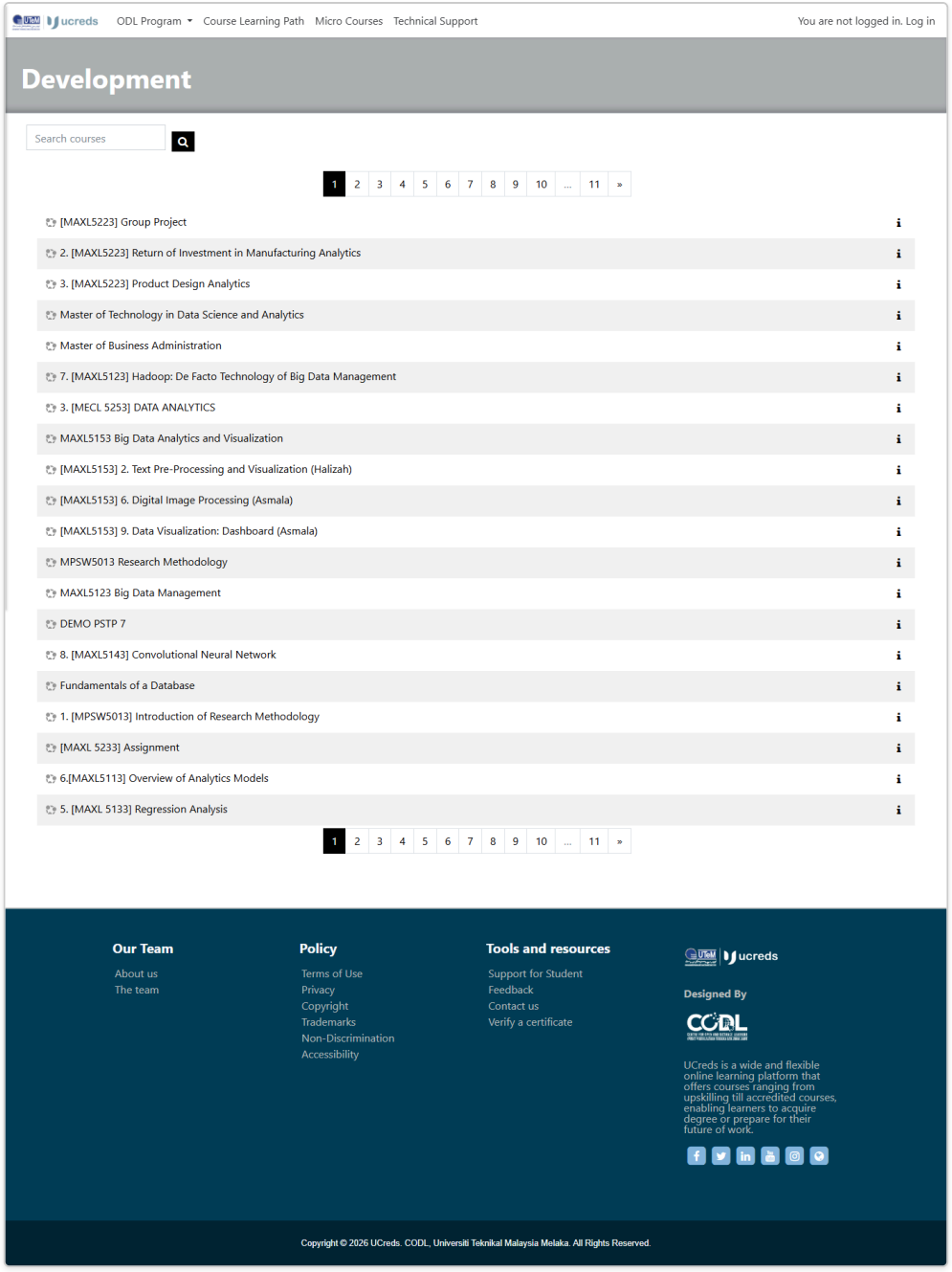

Figure 16: *Underlying Moodle Course Directory (course/index.php) displaying 214 Module Instances (`assets/11_moodle_course_index.png`)*

7.3 Flagship Course Overviews & Pedagogical Outlines

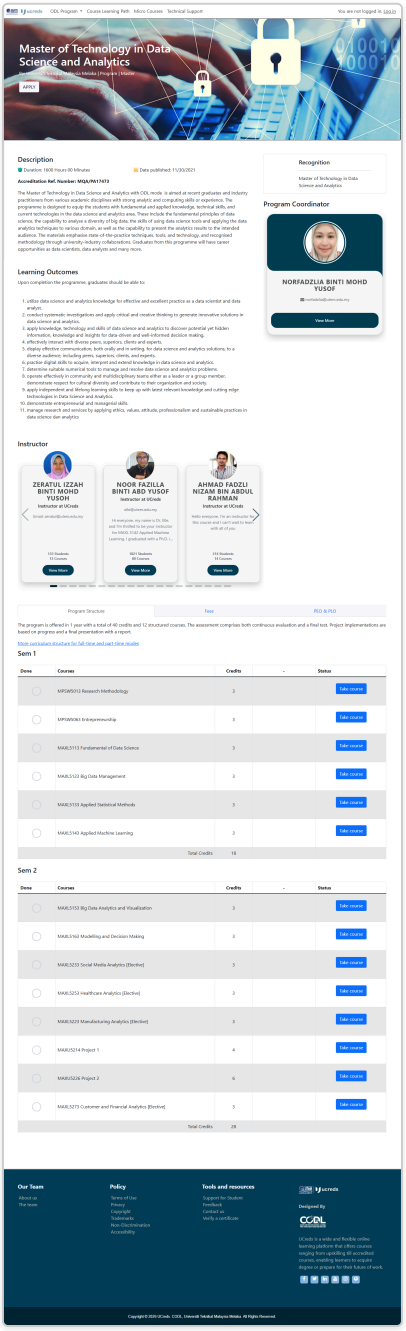

Figure 17: *Master of Technology in Data Science and Analytics (ODL) Overview with MQA/PA17473 Accreditation (`assets/course_080_mtdl_overview.png`)*

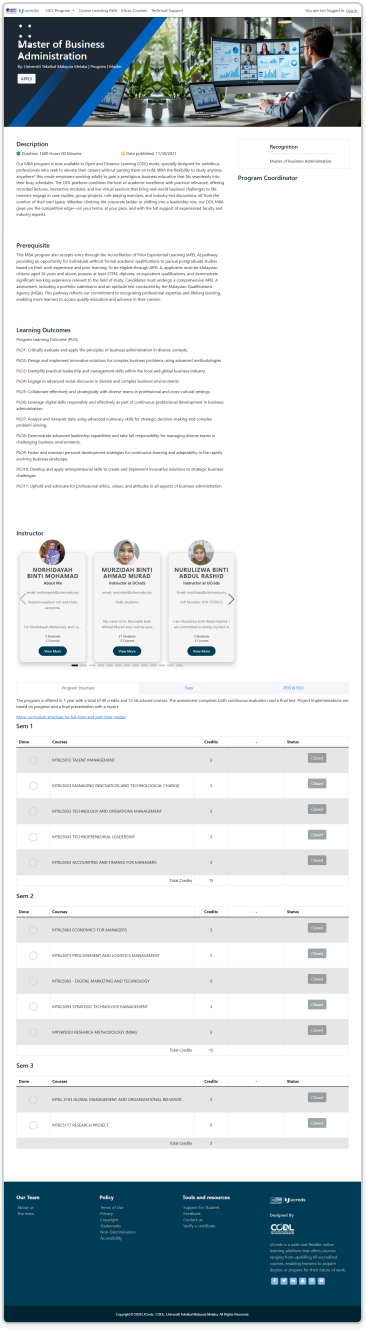

Figure 18: *Master of Business Administration (ODL) Overview displaying Programme Outcomes and Curriculum (`assets/course_078_mba_overview.png`)*

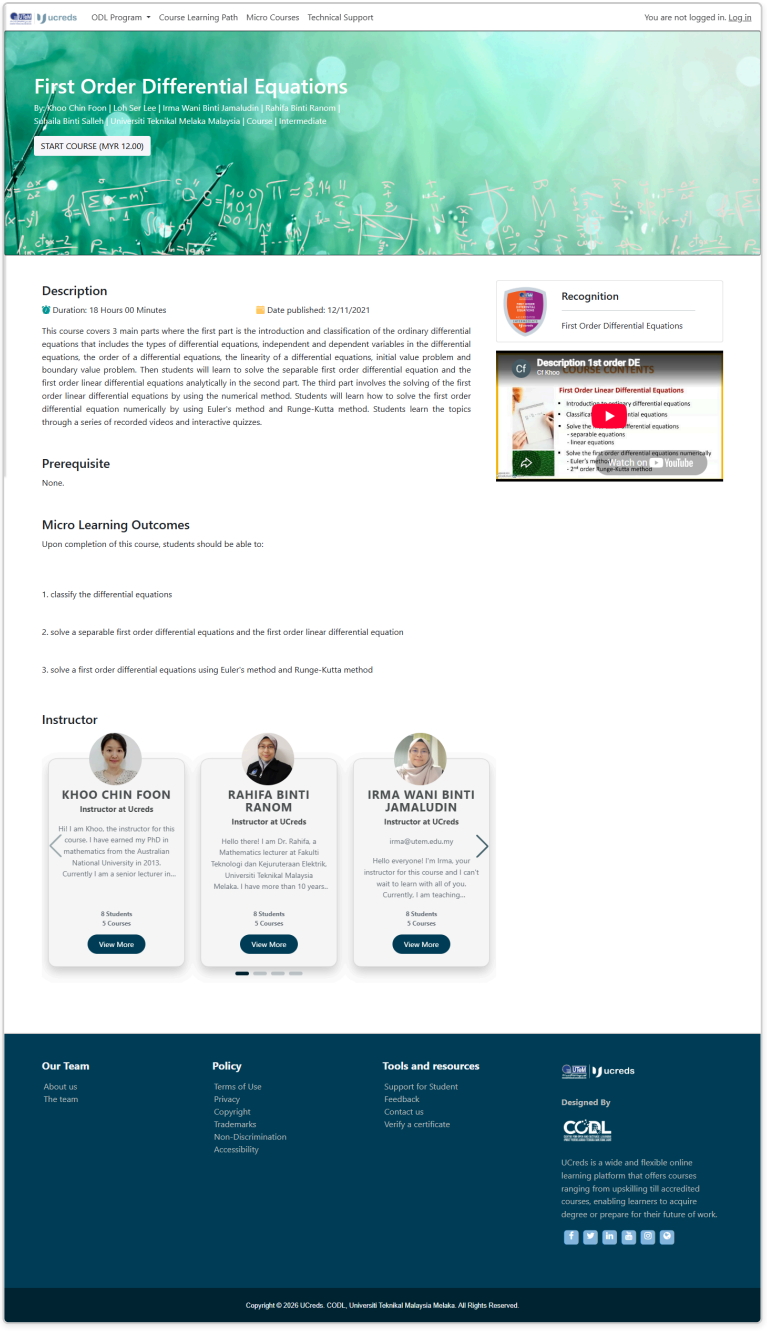

Figure 19: *First Order Differential Equations (ID 155) Course Overview showing Author Team and MLOs (`assets/course_155_math_ode.png`)*

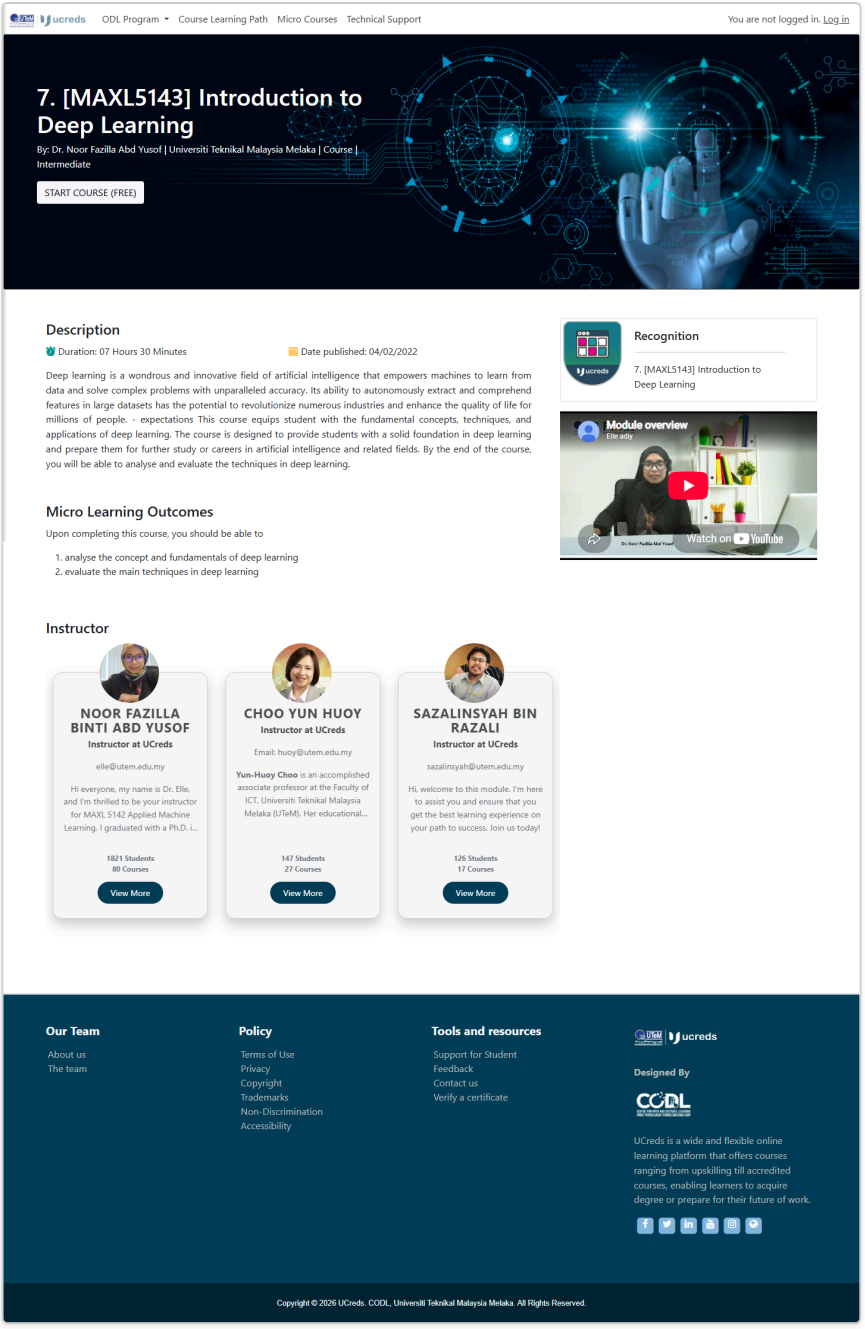

Figure 20: *Introduction to Deep Learning (MAXL5143 - ID 517) Course Overview with Syllabus Details (`assets/course_517_ai_deep_learning.png`)*

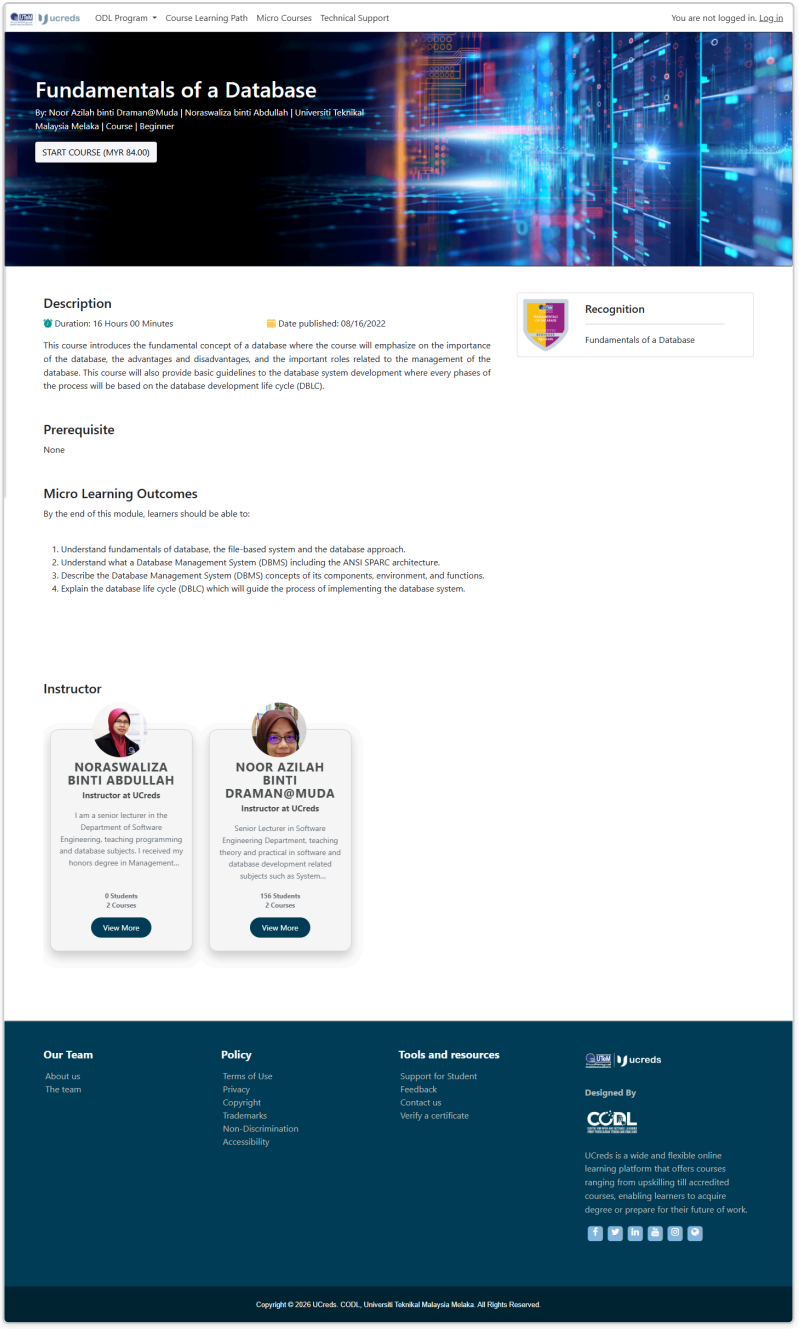

Figure 21: *Fundamentals of a Database (ID 644) Course Overview showing Learning Outcomes (`assets/course_644_database_fundamentals.png`)*

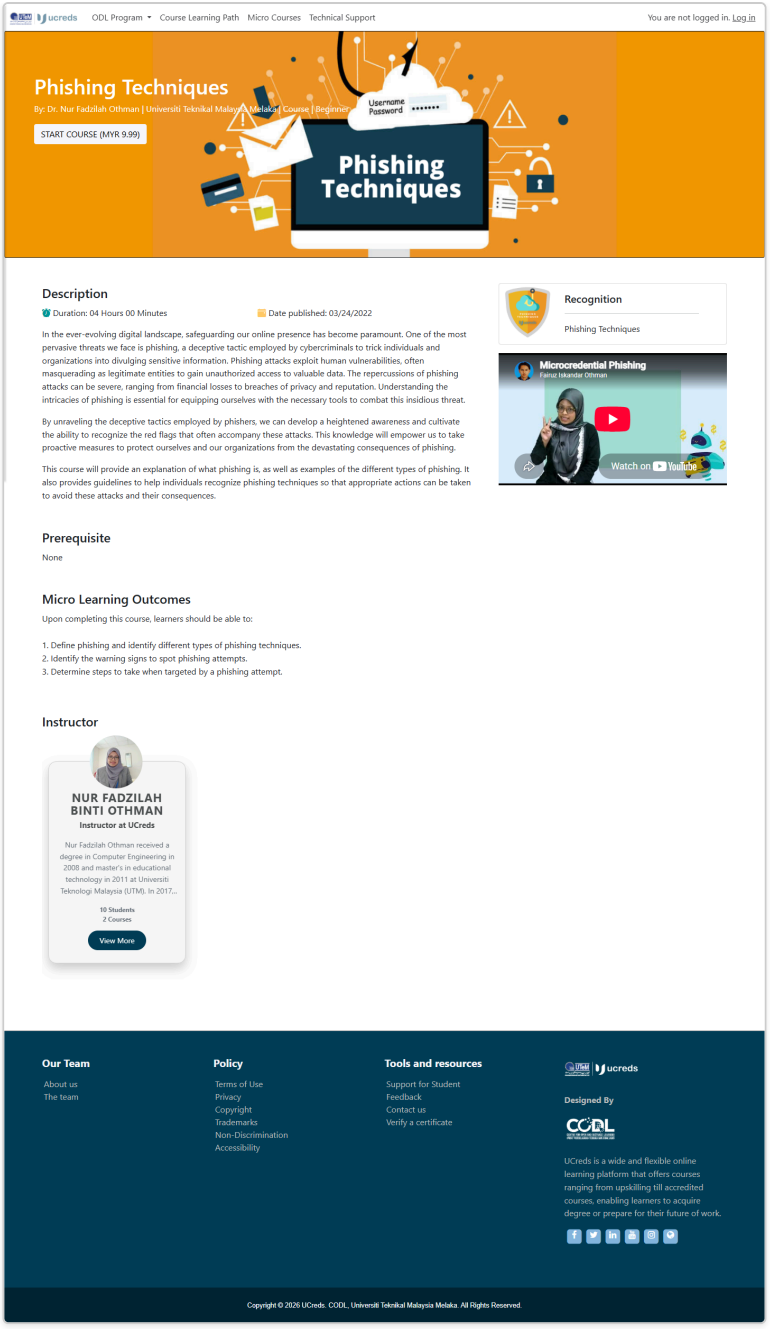

Figure 22: *Phishing Techniques (ID 486) Cyber Hygiene Course Overview (`assets/course_486_cybersecurity_phishing.png`)*

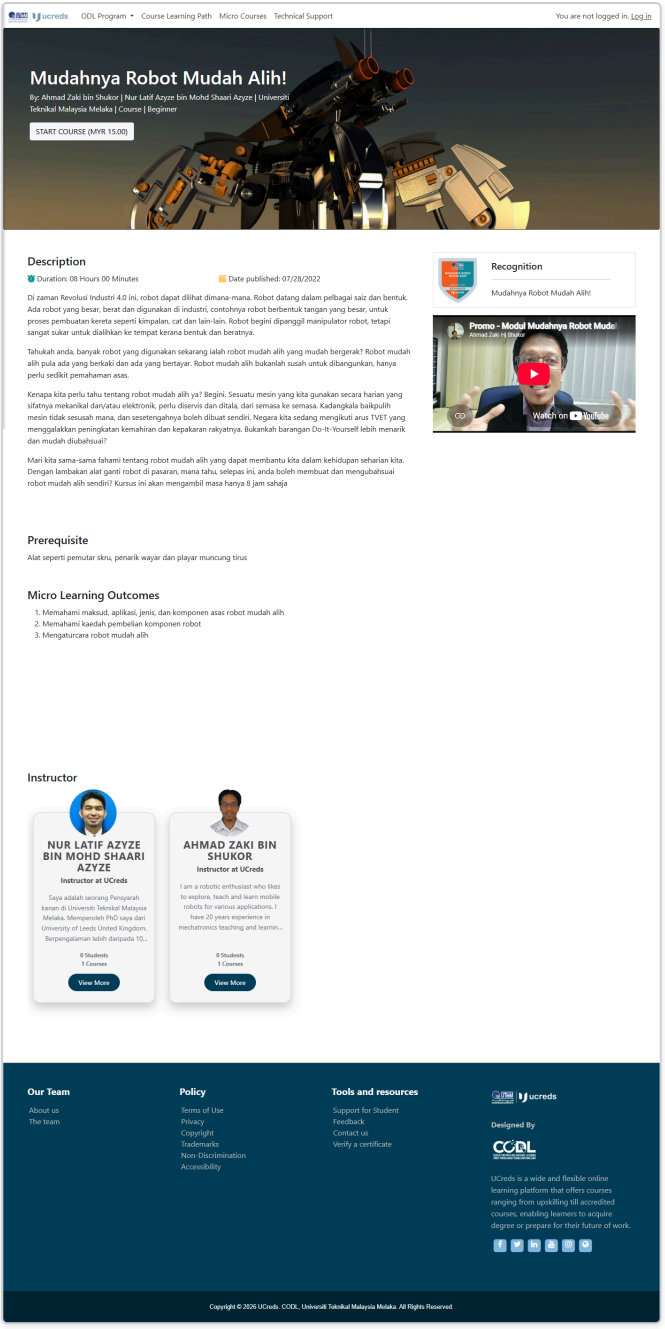

Figure 23: *Mudahnya Robot Mudah Alih! (Mobile Robotics - ID 573) TVET Course Overview (`assets/course_573_robotics.png`)*

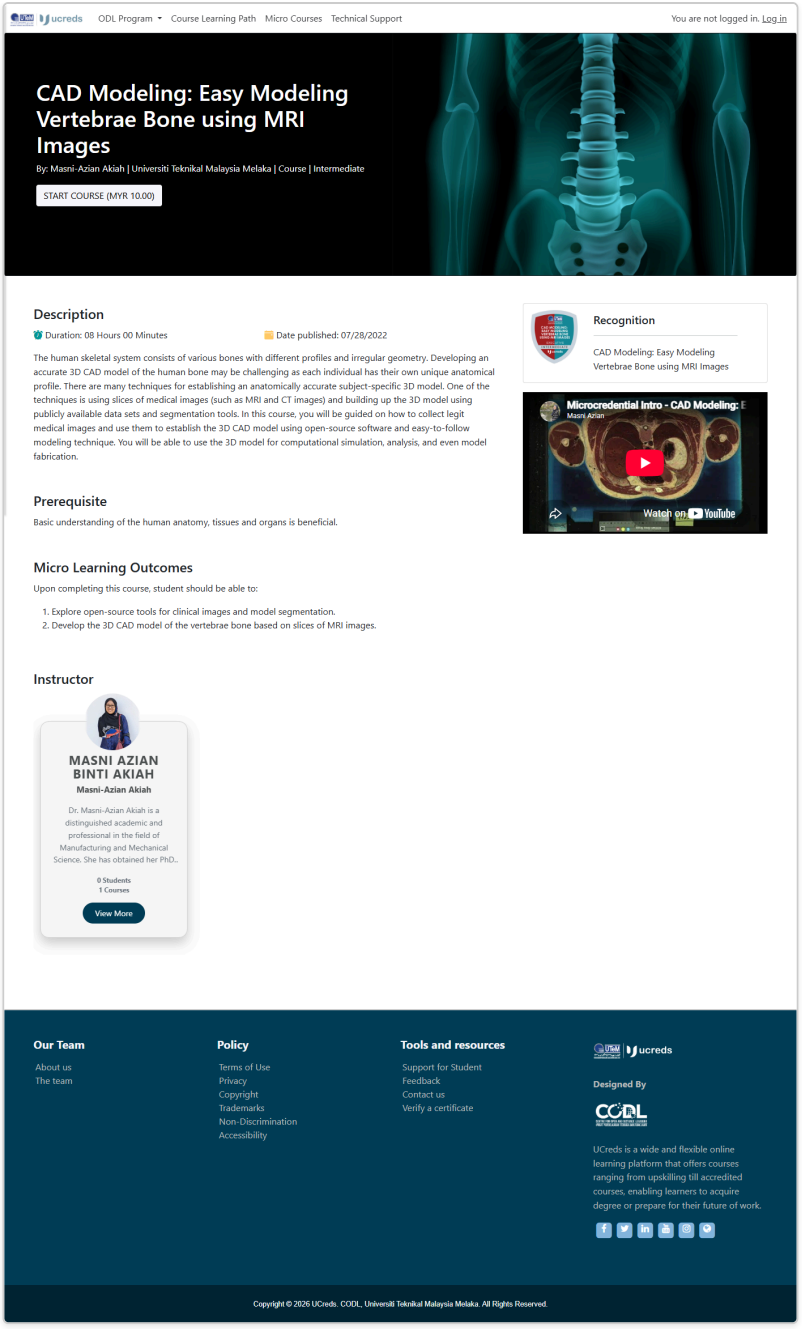

Figure 24: *CAD Modeling: Vertebrae Bone using MRI Images (ID 580) Course Overview (`assets/course_580_cad_vertebrae.png`)*

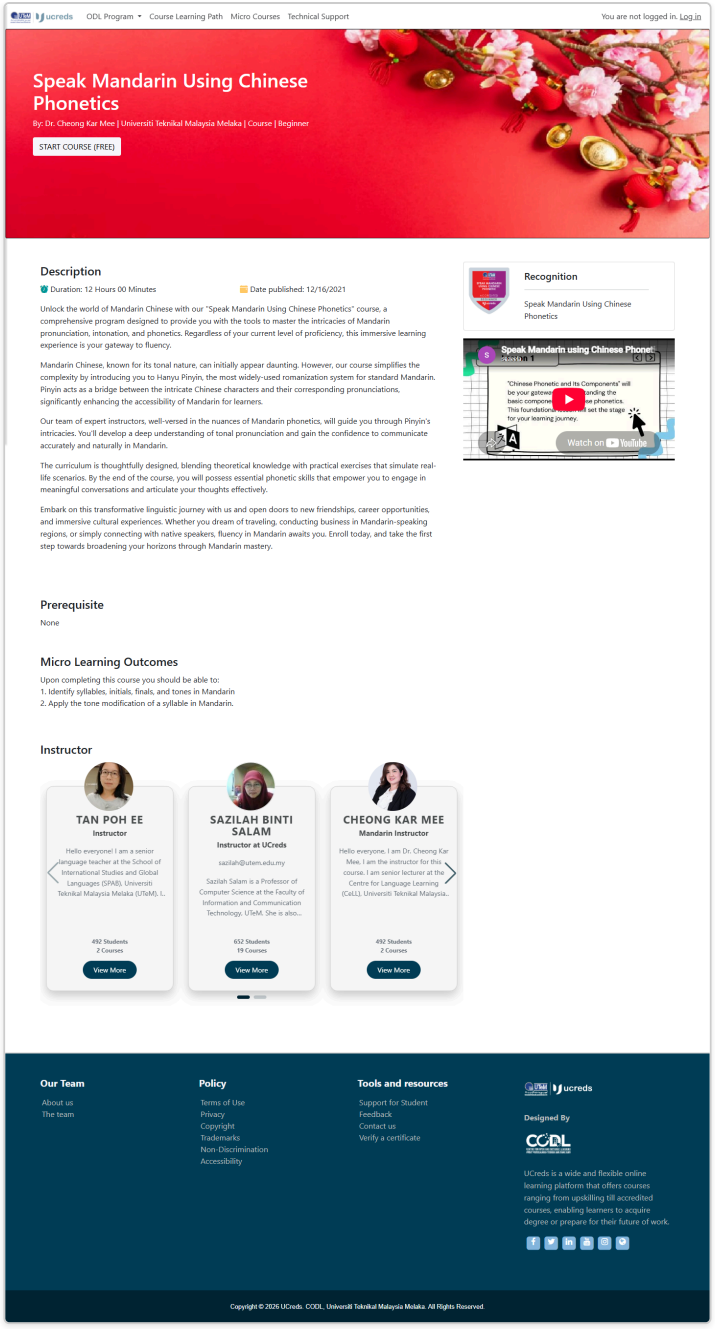

Figure 25: *Speak Mandarin Using Chinese Phonetics (ID 182) Language Course Overview (`assets/course_182_language_mandarin.png`)*

7.4 Student Support Knowledge Base & Institutional Guidance

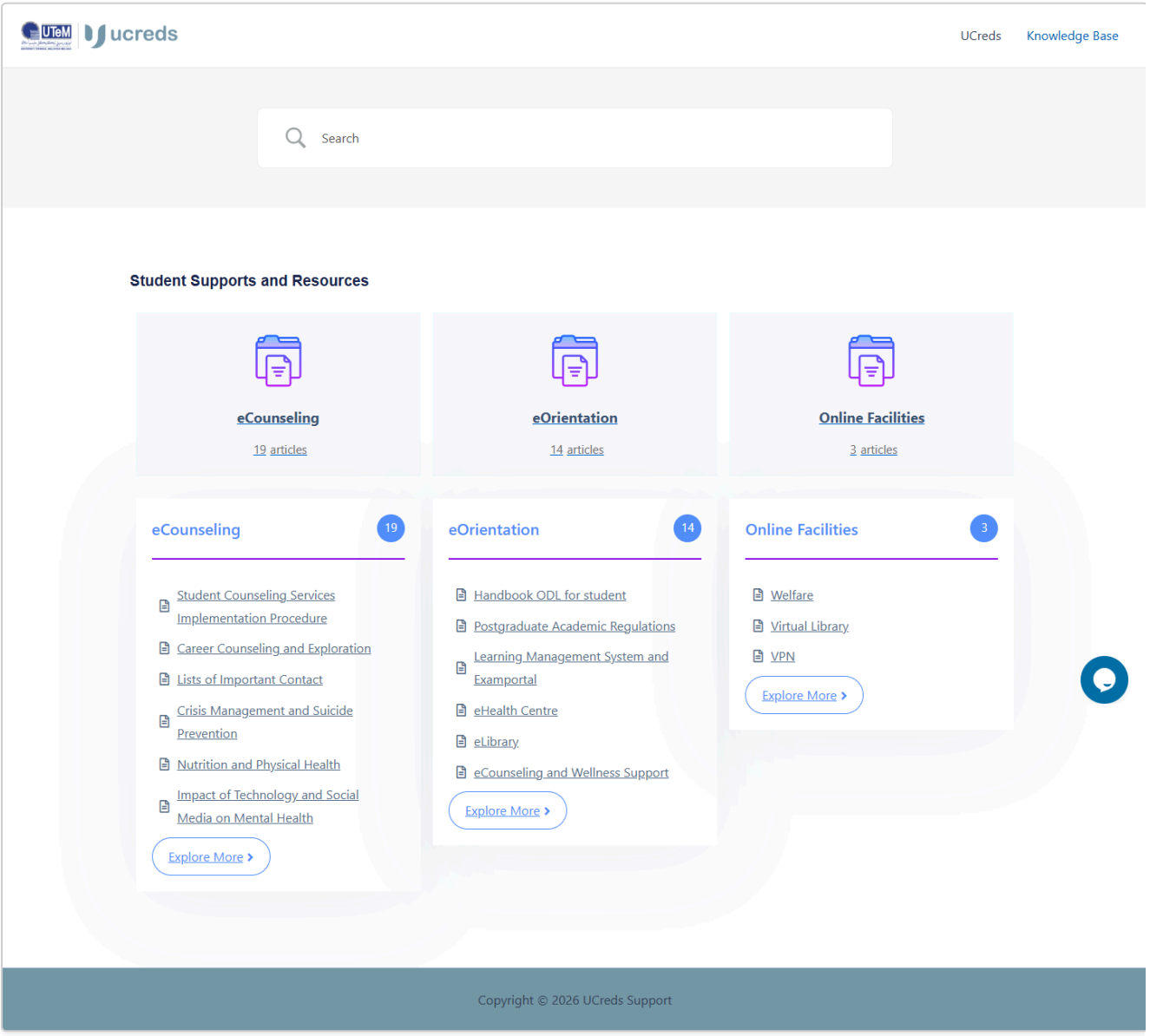

Figure 26: *BetterDocs Student Support Knowledge Base Portal featuring eCounseling, eOrientation, and Facilities (`assets/09_student_support.png`)*

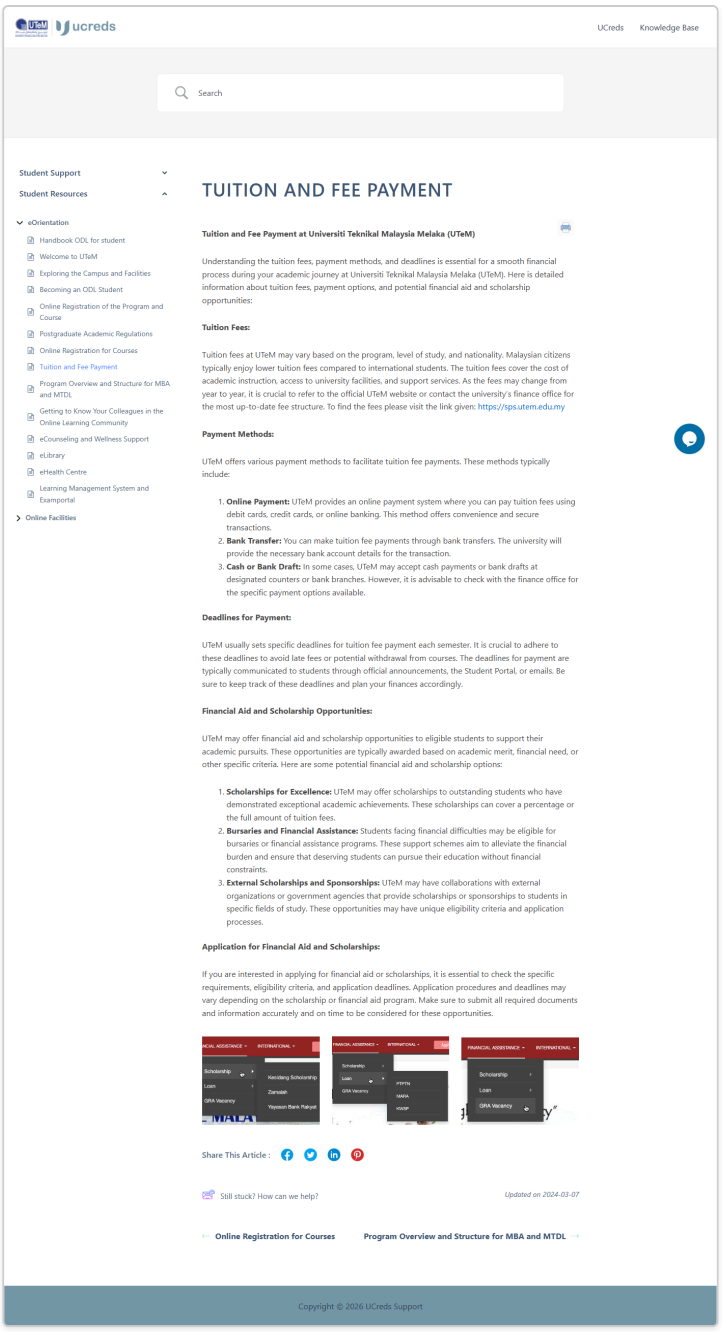

Figure 27: *Student Support Policy Article: Tuition and Fee Payment Schedule and Regulations (`assets/support_tuition_fees.png`)*

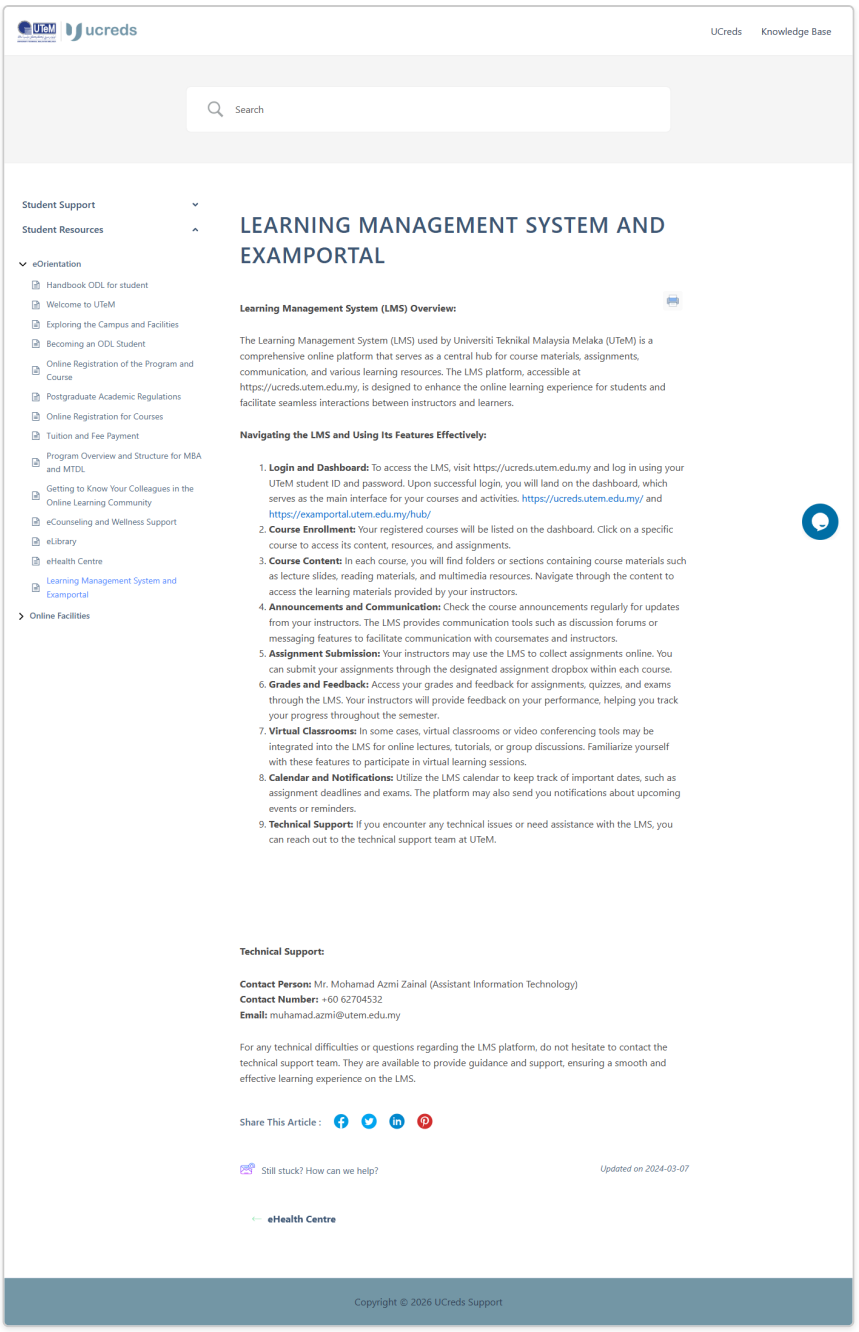

Figure 28: *Student Support Technical Article: Learning Management System and Examportal Guidelines (`assets/support_lms_examportal.png`)*

8. Strategic Alignment & Integration with UTeM MCMS

8.1 The Architectural Dichotomy: UCreds vs. MCMS

A critical outcome of this audit is clarifying the distinct institutional roles played by UCreds and the upcoming **Micro-Credential Management System (MCMS)**:

- **UCreds (The Learning Platform Gateway):** UCreds is the customer-facing Learning Management System (LMS) and storefront. Its responsibility is content delivery, learner interaction, automated quiz evaluation, forum discussions, and digital badging.
- **MCMS (The Authoritative Credential Registry):** MCMS is the central academic governance engine. It manages curriculum approval workflows (Faculty Deputy Dean & Senate), enforces the *Single Redemption Principle*, verifies quality standards (UMC vs SAMC), orchestrates graduation credit transfer, and deposits certified Academic Credit Records (ACR) to the national Academic Bank of Credit (ABC).

8.2 Data Entity Mapping

To ensure seamless interoperability between UCreds and MCMS, UCreds data objects map directly into the core entities specified in the MCMS Software Requirements Specification (SRS v1.0) and Software Design Document (SDD v1.0):

UCreds Concept	MCMS Domain Entity	SRS / SDD Reference	Integration Mechanism
Catalog Course / Programme	Offering (`MD-04`)	SRS 3.2.4 / SDD 4.4	Mapped via unique Course ID and Subject Code (e.g. `MAXL5123`).
Course Enrolment / Run	Cohort Run	SRS 3.2.4	Synchronized per academic intake semester.
Final Learner Result	Result Record	SRS 3.2.5 / SDD 4.5	Exported via batch CSV exchange (`JB-02`) or automated webhook.
Digital Badge / Pass	Credential (`MD-05`)	SRS 3.2.5	Issued in MCMS as a signed digital credential upon verification.
Moodle Core Engine	LearningPlatformGateway (`MD-07`)	SDD 4.7 / SRS 3.2.7	Adapter pattern supporting CSV and REST interfaces.

Table 12: *Data Model Mapping between UCreds and MCMS*

8.3 Automated Synchronization Architecture

MCMS incorporates scheduled background jobs designed specifically to ingest and reconcile learner outcomes from UCreds:

1. **Job `JB-02` (LMS Grade Sync):** Periodically polls the UCreds completion API or processes automated CSV gradebook exports to update student progression status in MCMS.
2. **Job `JB-03` (ABC Deposit Dispatcher):** Identifies eligible completed credentials that students have opted to deposit into the Academic Bank of Credit, signs the Academic Credit Record (ACR) with institutional keys, and transmits the payload to the national depository.

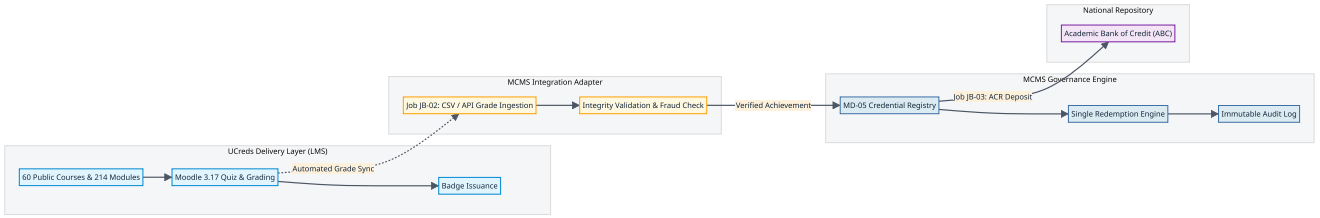

Figure 29: *Integrated Data Pipeline: From UCreds Course Delivery to MCMS Academic Depository*

8.4 Conclusion and Strategic Recommendations

This comprehensive audit affirms that UCreds represents a highly mature, academically robust, and technically well-architected digital learning ecosystem. To maximize its impact and accelerate integration with MCMS, the following strategic actions are recommended for the Centre for Open and Distance Learning (CODL):

1. **Formalize REST API Endpoints:** Transition the current batch CSV grade export to standard RESTful webhooks in Moodle (utilizing core web services with token authentication), allowing MCMS to ingest completion events in near real-time.
2. **Unified Taxonomy Standard:** Standardize course naming across Moodle so that every micro-course metadata record explicitly references its parent subject module code (e.g., `[MAXL5123]`) and SLT hours.
3. **Automated APEL & Credit Transfer Portals:** Surface a direct link on each UCreds course overview page enabling learners to preview how their earned micro-credential stacks into UTeM's degree programmes via the upcoming MCMS public portal.